

ORIGINAL ARTICLE

Causal network topology analysis: Characterizing causal context for risk management

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Abstract

The ways that risk assessments are commonly performed in organizations have limitations that undermine their quality. They typically focus on individual risk events one at a time but are weak at integrating their relevant causal context, into decision-making processes. Network topology analysis has previously been applied to address this weakness through quantitatively characterizing the importance of the causal interactions of risk events. However, there remains a lack of both clarity and consistency in terminology, methods, and interpretation of the results of this approach. This paper presents and formalizes causal network topology analysis, a methodology that contributes to (1) characterizing the causal context of a risk event to inform its management, (2) articulating the ontological concepts underpinning a repeatable topology network analysis, and (3) justifying the selection and usage of network metrics for this purpose. The theory and methodology are discussed, and an exemplar application to a mining project feasibility study is presented.

KEYWORDS

Causality, Context, Network Analysis, Risk assessment, Risk Management

1 | INTRODUCTION AND LITERATURE REVIEW

Risk management is embedded in all organizations. It is used to inform decision-making, in fields as diverse as health and safety, crisis management, community relations, and environmental management. However, weaknesses in risk management in general, and risk assessment in particular, have been recognized. One key facet that is consistently challenging to organizations is the rich appreciation of *context* for a system undergoing risk assessment. The need for taking into consideration both the external and internal context of a system is highlighted in ISO31000 (ISO 2018); however, this is challenging because context is multidimensional, and uncertainty is generated in the risk management process through a *lack* of integration of information: not seeing the full picture from the multiple “pieces of the decision-makers’ puzzle.” One way forward is by improving the *integration* of information across the *causal* context.

Considering the causal context for risks is valuable when it is concurrent with controlling each risk (Dekker et al., 2011). In any organization, risks are causally interconnected, and it is often the combination of multiple failures that lead to a catastrophe in a complex system (Cook, 1998). However, conventional risk management practice lacks an integrated approach to risk identification. Most risk management strategies seek to optimize the management of each risk individually, but risk events do not exist in a vacuum: the consequences of one risk may cause others, or the controls for one risk may prevent other risks. In the hazard and operability (HAZOP) study, the focus on deviations may distract attention from identifying causes, which are sometimes more critical (Baybutt, 2015). Similarly, failure mode and effect analysis, and failure mode effect and criticality analysis, identify potential failures and their causes and consequences individually (Ben-Daya, 2009; Borgovini et al., 1993).

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TABLE 1 Main approaches using graph and graph theory to model causal scenarios.

Method	Key reference(s)
Domino theory	Ferry (1988)
Swiss cheese model	Reason (1997)
Accimap	Rasmussen (1997); Rasmussen and Suedung (2000)
Why–because graph	Gerdsmeyer et al. (1997); Ladkin and Loer (2001)
Fault tree analysis (FTA)	IEC (2008); Khan and Husain (2001); Vesely et al. (1981)
Event tree analysis (ETA)	Ericson (2005); IEC (2011); Vélchez et al. (2011)
Bow-tie analysis	Cockshott (2005); Pasman (2015a)
Fishbone/Ishikawa diagrams	Juran et al. (1999)
Bayesian networks	Abimbola et al. (2015); Khakzad et al. (2013b); Pasman (2015b); Pearl (1985); Xin et al. (2017)
Functional resonance analysis method (FRAM)	Hollnagel (2012); Rosa et al. (2015)
Systems theoretic process analysis (STPA)	Ishimatsu et al. (2014); Leveson and Thomas (2018)
Failure environment analysis tool (FEAT)	Pack and Wadsworth (1994)

Causal scenario¹ identification and description are critical activities in both accident investigation and risk assessment (Cameron et al., 2017). The use of graphs and graph theory to model causal scenarios in process safety has existed for many years, for the purpose of identifying possible failures, detecting and predicting abnormal events, and performing risk assessment. Some of these main approaches are shown in Table 1.

These approaches, firstly, identify *discrete* units of study, such as *events* (FTA, ETA, Accimap, FEAT), *functions* (FRAM), system *components* (STPA, Swiss Cheese), or accident *contributing factors* (Domino model, Ishikawa/Fishbone diagram). These discrete units are then connected via *links* that represent physical, functional, or causal relationships. Other approaches that combine different graphical methodologies in Table 1 include:

- Dynamic risk assessment: (1) Dynamic risk and safety analysis using FTA (Čepin & Mavko, 2002), Bow-tie (Abimbola et al., 2014), Bayesian network (Adedigba et al., 2016; Chen et al., 2021) to update risk values and risk profile through the lifespan of a process; (2) Domino effects analysis, which applies Bayesian networks to represent propagation pathways, predict and update propagation occurrence probability (Khakzad et al., 2012), to reduce the escalation probability of domino effects (Khakzad et al., 2014), and quantify firefighting systems performance along with interdependencies between events in temporal evolution during fire escalation (Khakzad et al., 2017);
- Causality-based accident modeling techniques, including system hazard identification, prediction, and prevention, combine the Bayesian mechanism with ETA and FTA to update and predict abnormal events occurrences (Rathnayaka et al., 2011a, 2011b);

- Data-driven analysis using the Bayesian network to investigate the impact of risk factors on railway safety (Liang et al., 2017; Shi et al., 2024), to simulate risk scenarios in offshore drilling operation (Chen et al., 2021), and to identify and quantify interrelationships between hazards (Bouzembrak & Marvin, 2019) and the vulnerability (Marvin et al., 2020) in supply chain systems;
- Analyses that map Bow-tie (Khakzad et al., 2013a), transfer FTA (Bobbio et al., 2001), encompass domino theory (Khakzad et al., 2012), and convert interpretive structural modeling (Sajid et al., 2017; Wu et al., 2015) into Bayesian networks for addressing the limitations inherent in these individual approaches.

However, all these past approaches do not take advantage of the extra information stored in the *topology* of the network structure of the model itself; namely, the structure of how the links between the discrete units of study, called *nodes*, are distributed. This could limit the ability of those approaches to fully capture the complex nonlinear relationships between causal factors, as Qiu et al. (2021) stated:

Although existing studies have analysed the accident-causing mechanism from a systematic perspective, they still cannot fully reflect the hierarchy and correlation among accident-causing factors, nor the complex non-linear interactions between causative factors.

Network topology contains information that can be extracted by calculating a variety of network metrics (Boccaletti et al., 2006). A number of previous authors have applied network topology analysis to characterize the causal context of past incidents and have identified the potential value gained from this approach. Zhou et al. (2014) constructed the Subway Construction Accident Network to characterize the causal relationships between accident events, the accident propagation rate and propagation ability. Thoroman and

¹ Causal scenario: The causal context of all events in that scenario. For example, each line of a risk register can be seen as a causal scenario (a risk with its causes, consequences, and controls). In the network context, a causal scenario can be represented as a subgraph of a causal network.

Salmon (2020) applied network topology analysis to near-miss incidents, to reveal the interactions between factors within incidents, and quantify the influence of each factor for comparison. Sun et al. (2020) combined the quantitative risk analysis and network topology analysis for application to a chemical process system to quantify pandemic-related impacts on the chemical industry. Other examples of the applications of topology analysis using networks across fields and purposes can be found in the literature (Gao & Ren, 2013; Huang et al., 2020; Khakzad & Reniers, 2015; Klockner & Toft, 2015, 2018; Lam et al., 2019; Ma et al., 2022; Qiu et al., 2021; Seligmann et al., 2018, 2019; Yang et al., 2023; Zhou & Irizarry, 2016; Zhou et al., 2021), demonstrating that network topology analysis has been previously identified as suitable for characterizing the causal context of complex systems.

Two main claims have been made about the application of topology analysis for this purpose: The first claim is that it enhances the understanding of accidents. For example, identifying the most important causal factors within railway (Huang et al., 2020; Zhou & Irizarry, 2016; Zhou et al., 2014), coal mine (Qiu et al., 2021), chemical (Seligmann et al., 2018), or ship grounding accidents (Ma et al., 2022), to prevent reoccurrence. The second claim is that the use of topology metrics can inform risk treatment strategies. This includes using network robustness analysis (Boccaletti et al., 2006) on railway scenarios or financial risk scenarios (Gao & Ren, 2013; Zhou et al., 2021), and on a HAZOP study related to a heat exchanger start-up process (Seligmann, 2021).

Despite all this, there are important gaps that remain: (1) *Terminology*: There exists a diversity of concepts used to build and analyze networks in previous works. Examples include differences in definitions of network nodes and edges, and differences in how edge weights are determined. This diversity creates ambiguity in what these networks actually represent. One reason for this is that the set of concepts, or ontology, used by previous authors to convert their raw data to networks for subsequent analyses is not explicit. This makes it difficult to apply these network analysis methodologies to new problems. (2) *Interpretation of network metrics*: Past literature has applied various network topology metrics for multiple purposes; however, the definition and the justification of the choice of network metrics used are inconsistent, as stated by Morrison et al. (2022): “The diversity allows network analysis to be applied across myriad research domains and contexts.... Challenges arise in identifying which metrics to choose.” Furthermore, despite the additional insights claimed by using these metrics, examples embedding them into the process of risk management are rare. Those examples which do exist (e.g., Peeters et al., 2022; Sun et al., 2020) share limitations with the literature cited above.

To conclude, more rigor is required for network topology analysis to be systematically and consistently applied to support the risk management process, in order to characterize the causal context of risk events and better integrate information for subsequent decision-making.

Therefore, we present an approach named Causal Network Topology Analysis (CaNeTA), which:

1. Articulates the network topology analysis so that it is repeatable for multiple risk management problems.
2. Defines a set of concepts used to extract and combine data to form the networks.
3. Provides guidance on how to justify the *selection* of which network metrics to use, for particular patterns or questions.
4. Interprets how metric results can be used for characterizing the causal context, thus informing risk management decision-making.

In this paper, we build on the previous work of Seligmann et al. (2019, 2018) and address the four points above in the following sections: Section 2 presents the CaNeTA methodology, provides a *formalized* description and *definitions* of the underlying concepts, and *justifies* the choice of metrics used by providing an interpretation of what they mean when applied to a causal network. Section 3 presents a case study illustrating how CaNeTA has been applied to a mining project feasibility study and includes *guidance* on how to interpret the results. Section 4 discusses the contributions of CaNeTA to previous causal network topology analyses; Section 5 contains conclusions and limitations of this study and describes future work.

2 | CAUSAL NETWORK TOPOLOGY ANALYSIS

The CaNeTA methodology is illustrated in Figure 1. This approach enables the consistent application to various datasets, allowing the characterization of causal context for risk assessment. Currently, it consists of four stages: (1) data preprocessing, (2) causal network development and visualization, (3) topology metrics calculation, and (4) metric interpretation.

2.1 | Data preprocessing

To prepare risk data for CaNeTA, we search for events and their causal links within the input data, such as risk registers. Data are then transferred into a tabular format suitable (i.e., the input data are discretized) for conversion to an adjacency matrix and to a causal network using our Python-based software. The relative position of events in these tables specifies their causal relationships. Event information can be collected from various data sources because of the fundamental definitions we use for the concepts of *event* and *cause*. Here we extend upon the ontology described in Seligmann (2011) to define these terms.

Event: A change in one or more *properties* beyond some *threshold*, embedded in *space* and *time*. A property

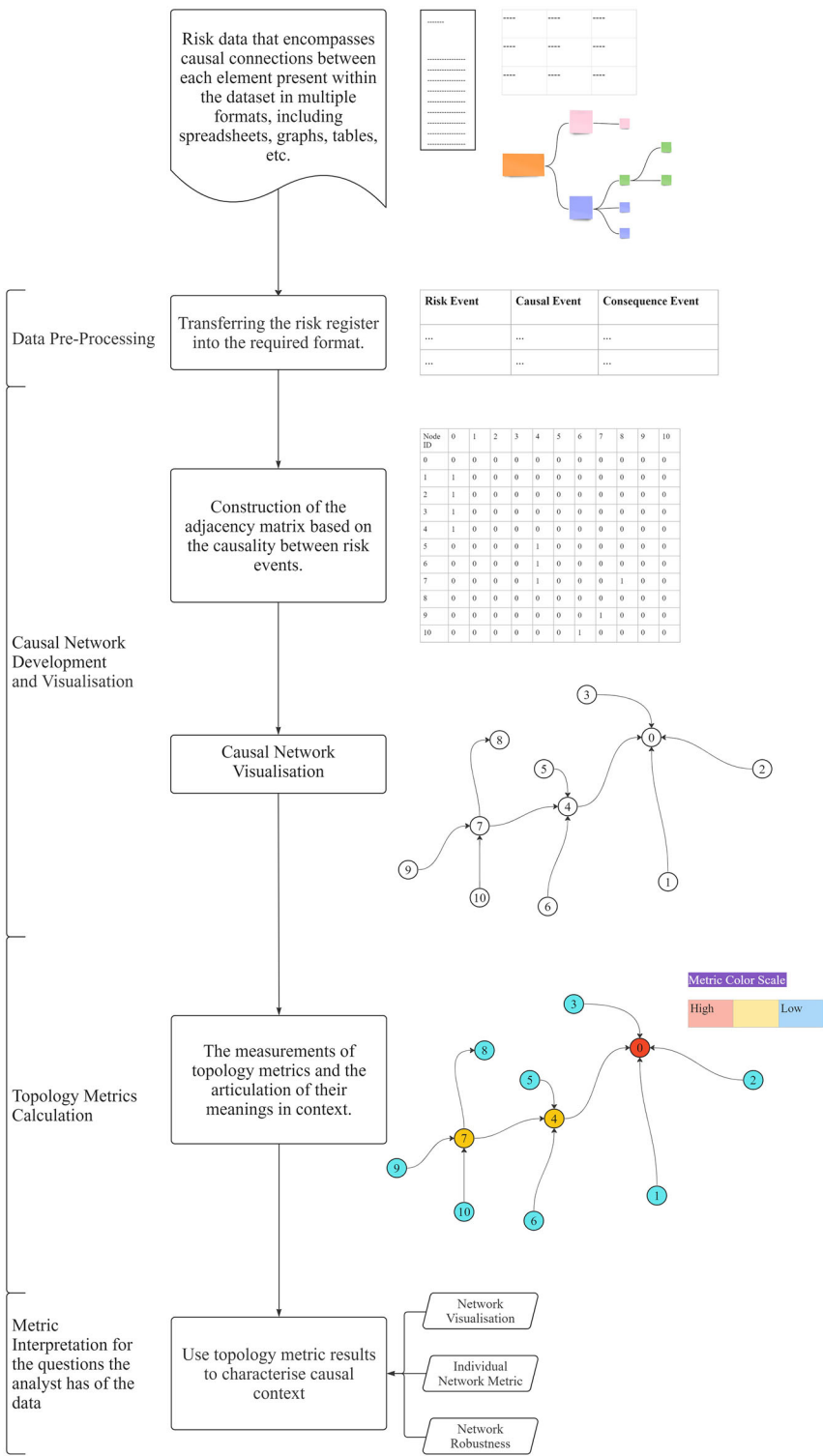


FIGURE 1 Causal network topology analysis (CaNeTA) methodology.

is a feature of the system of interest, where the set of properties form its state. An example of an event is “excessive increase pressure in pipe,” a typical risk event from chemical process safety. The property is *pressure*. The term “excessive increase” implies a *change* in the value of pressure beyond some accepted *threshold*. This change will occur over a finite *time* period, even if it is very short. “Pipe” denotes the loca-

tion of the event, being associated with a region of *space*.

Cause: A necessary and sufficient condition for an event to occur. A cause is itself a set of events, containing at least one element.

To date, the type of data used for CaNeTA are text descriptions of events captured in tables. For example, with a risk

register, there is often a column named “risk event,” and another named “consequences.” The names of these columns imply a causal relationship between the events. We take advantage of this embedded causal information to build the causal network. Any data that fit the definition of an event given above can be included in the CaNeTA input, and this is thus not limited to text captured in tables. For example, events can be identified from time series (Corrêa et al., 2022), from accident reports (Huang et al., 2020; Ma et al., 2022; Qiu et al., 2021; Seligmann et al., 2018; Thoroman & Salmon, 2020; Zhou & Irizarry, 2016; Zhou et al., 2021, 2014), and from safety occurrence reports (Klockner & Toft, 2015).

2.2 | Network development and visualization

Once the input data are discretized and put into the correct format, these are converted into an adjacency matrix (Boccaletti et al., 2006), in order to visualize it as a network. A network consists of nodes and edges, where the edges link the nodes to form a structure (Boccaletti et al., 2006; Newman, 2003). A causal network is modeled here as a directed, weighted graph, describing the causality between events within the scenario (Seligmann et al., 2018, 2019). Considering a directed graph G , with N representing the nodes. If there is a causal relationship between two nodes N_i and N_j , an edge will be established from node i to node j with a numerical edge weight, W_{ij} . A matrix representation is constructed to describe the network. This adjacency matrix, a , has the same number of rows and columns as the number of nodes, where each entry, a_{ij} , is the weight of the edge linking node i to node j , W_{ij} (Boccaletti et al., 2006).

In CaNeTA, nodes represent events. This is distinct from the labels used in risk registers where events are described as, for example, causes, risk events, or consequences. Such labeling is useful for sequencing, indicating causal dependencies when doing risk identification, but here they are all treated as events. This removal of labels, when the network is formed, has the advantage of breaking down the barriers immediately between contexts in the input data, showing the links between all events, enabling the representation and analysis of the causal context. These nodes are linked by edges with weights on them. The weights are determined by applying the widely-used AND and OR gate operator (Chybowska et al., 2021; Stevenson et al., 1991; Seligmann et al., 2018, 2019), applied as a modified version of the approach from Seligmann et al. (2018). Figure 2 shows three examples of how these logic gates are applied to describe causal links between events. An OR gate example illustrates the causal relationships between an event and its single cause. For instance, “High temperature in the pipe” causes “High pressure in the pipe.” The causes within an OR gate are independent of each other. An AND gate example that articulates the causal relationships between an event and its causes may be, “High Temperature in the Pipe” and “Cooling System Failure” causes “Overpressure in the Pipe.” Those compound events formulate a necessary and sufficient condition for their consequences to occur. Edges in an OR gate all have a weight of 1, whereas edges in an AND

gate will have edge weights that reflect the number of causes in the gate: 2 causes, each have edge weights of 2; 3 causes, each have an edge weight of 3, etc.

The work we are presenting in this paper is not primarily concerned with calculating probabilities of events. We assert that useful insights can be gained from studying the topology of causal networks for risk management applications, separately from calculating the probabilities of events within causal networks. Therefore, we consider that AND/OR gates are sufficient for representing the essential causal relationships between events, specifically because we are interested in the topology of these causal relationships. This approach may not be strictly true because even the most assertive natural language expression for causality is subject to exceptions (Pearl, 2009). However, the likelihood data of risks or causes can also be subjective. Therefore, we have simplified the probability distributions of events that are normally used in other causal network approaches, for example, the causal Bayesian network (Pearl, 2009). If these events/nodes are within the network, then they will have a possibility to occur. If any event/node has been removed, then they will not happen in this scenario (which will be further described in Section 2.4). We do not overlook the likelihood and consequences data within the risk register. Instead, we integrate this information into CaNeTA during robustness analysis and compare results both with and without it, as described in Section 2.4.

The process of transforming part of a risk register (the top table) into a formatted table (the middle-right table), and then into an adjacency matrix that serves as the basis for constructing a causal network, is illustrated in Figure 3. Elements within the top table columns “Description of existing control” (first), “Causes” (second), “Description of the risk” (third), and “Impacts” (fourth) are separated as individual events. These events will be split and categorized into causes, consequences, and risk events, to aid in the transfer from risk register formats. The reason behind this is that, if two events are within an AND gate and cause another event, then these two events together are treated as one cause. The events within the second, third, and fourth columns are matched by the corresponding controls, which are described as “AND GATES.” Normally, the “AND GATES” column would not appear in a risk register; here, we have this additional column for a clearer explanation. When the “controls column” in risk registers are brought into a causal network, the control descriptions are treated as control failures. The idea is that the controls in an AND gate need to fail to let the subsequent event occur. The status of these controls (e.g., failure of) will be added before them to form an AND gate, with risks or causes. This process is repeated for all elements within the risk register, and the output (the middle-right table) is converted into an adjacency matrix for topology analysis.

The first step of analysis after the network is created is *visualizing* it, as shown in Figure 4(a) and (b). Reflecting on the visual structure allows questions to be asked about potential missing events (nodes with red dotted lines) or potential missing causal links (edges with red dotted lines) connecting two events (nodes). Asking these questions of the visualized net-

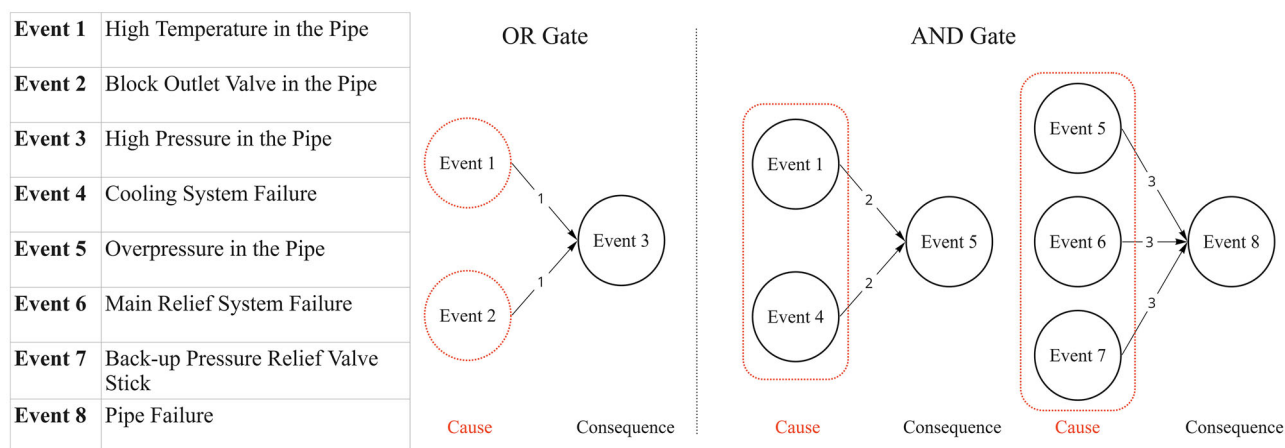


FIGURE 2 An example of a pipe system illustrates the logic gate that represents causal relationships between events (red dotted lines refer to what a cause is in each example).

work can also lead to insights about the quality of the input risk data.

2.3 | Topology metric calculation: Characterization of causal context

To characterize the causal context of the risk events, network metrics are calculated. Table 2 contains commonly used metrics that are applied to demonstrate the concept. The selected nine metrics are from past literature. We are using these metrics as examples for representing the CaNeTA methodology and providing interpretation of these metrics to justify where to use them in different situations. These metrics are applied to individual nodes i (N_i), representing individual events (N). The choice of network metrics used should reflect what the analyst believes is worth learning about the causal context. For example, to locate causally important events or to prioritize risks for subsequent treatment. The choice of which metrics to use affects what can be learned from the topology results.

The values of topology metrics indicate various types of information about the causal scenario. The most straightforward example is in the set of nodes with zero *in degree* and zero *out degree*. Figure 5 illustrates the sets of nodes on the outermost layer of the network (zero *in degree* and *out degree*), indicating the events that themselves have no causes or consequences, respectively, in this context. These sets of events can be used, for example, to check if the results of the risk assessment meet its designed scope and indicate if the search for causes and consequences has been pursued for enough up and downstream.

2.4 | Network robustness: Fragility of causal context

Network robustness is often demonstrated by node removal, structured either randomly or intentionally (Boccaletti et al.,

2006), and shows either robustness or fragility in a scale-free network (Albert et al., 2000). Node removal in this way is a model of the process of treating risks, an approach that is intended to interrupt the causal relationship between a risk event and its causes or consequences.

Robustness analysis in CaNeTA follows the logic gate in Figure 2, which characterizes the fragility of the causal context. When a single node is chosen to be removed, its incoming and outgoing edges will be removed as well. If the outgoing edges are within an AND gate, or within multiple AND gates, then all edges within these gates will be removed. Traditionally, robustness analysis via node removal targets highest *degree* nodes (Newman, 2003), but intrinsically any metric could be used.

As Newman (2003) stated, while removing the nodes with the highest value of the chosen metric, some would recalculate the metrics after each elimination, while others follow the initial calculated metric order. In CaNeTA, the metric values are recalculated after each node is removed. This is preferred because, after each node removal step, the resulting network is essentially a different network compared with the previous one. Consequently, relying on the original node metric values is not meaningful as the network has changed. As will be shown, the results can be used in a variety of ways. By breaking up the causal network, we reduce the maximum consequence associated with the original causal network to the maximum impact associated with the remaining part of the network. Therefore, performing network robustness analysis provides an “efficient” means to break down the size of the network—interpreted here as an idealized model of treating or preventing risks. The process can be made more efficient by structuring the order of node removal based on the topology metrics that have been calculated, with *efficiency* increased by choosing the topology metric that results in the minimum number of individual risks that are required to be prevented or treated to reduce the size of the network to zero. The number of nodes remaining after each removal step is plotted against the number of nodes removed to form the

TABLE 2 Causal network metric description.

Metric	Symbol and formula	Definition	Meaning
In degree	$d_{in}(N_i) = \sum_{N_j \in N} a_{ji}$ Here a_{ij} represents a link from node j to i, having a value greater than 0 in the adjacency matrix (Boccaletti et al., 2006).	The number of incoming edges from other nodes to node N_i .	The number of causally upstream ^a events involved in causing event N_i .
Out degree	$d_{out}(N_i) = \sum_{N_j \in N} a_{ij}$ Here a_{ij} represents a link from node i to j, having a value greater than 0 in the adjacency matrix (Boccaletti et al., 2006).	The number of outgoing edges from node N_i to other nodes.	The number of causally downstream ^b events that event N_i is involved in causing.
Degree	$d(N_i) = \sum_{N_j \in N} a_{ij} + \sum_{N_j \in N} a_{ji}$ (Freeman, 1978)	The sum of <i>in degree</i> and <i>out degree</i> of node N_i .	The number of total causal relationships event N_i has with other events, both upstream and downstream.
In strength	$s_{in}(N_i) = \sum_{N_j \in N} \frac{1}{w_{ji}}$ Here w_{ji} is defined as the entry in the adjacency matrix, representing the weight on that link, in a weighted directed network.	The sum of the reciprocal of incoming edge weights from other nodes to node N_i .	The weights quantitatively indicate the tendency of event N_i to be caused by other events, considering their logic of causality and the number of causes. The reason for using reciprocal is based on the following scenario: Suppose two edges within one AND gate are pointing to N_i , whereas another two edges within one OR gate are pointing to N_i simultaneously. There are logically more possible ways for N_i to be caused by relationships (edges) through the OR gate; therefore, the edge weights within the OR gate should be regarded as more significant.
Out strength	$s_{out}(N_i) = \sum_{N_j \in N} \frac{1}{w_{ij}}$ Here w_{ij} represents a link ejecting from node i to j together with the weight, in a weighted directed network.	The sum of the reciprocal of outgoing edge weights from node N_i to other nodes.	The weights quantitatively indicate the ability of event N_i to cause other events, considering their logic of causality and number of consequence events. The reason for using reciprocal is based on the following scenario: Suppose two edges within one AND gate are outgoing from N_i , whereas another two edges within one OR gate are outgoing from N_i simultaneously. There are logically more possible ways for N_i to cause other events by relationships (edges) through the OR gate; therefore, the edge weights within the OR gate should be regarded as more significant.

(Continues)

TABLE 2 (Continued)

Metric	Symbol and formula	Definition	Meaning
Strength	$s(N_i) = \sum_{N_j \in N} \frac{1}{w_{ij}} + \sum_{N_j \in N} \frac{1}{w_{ji}}$ (Boccaletti et al., 2006)	The sum of <i>in strength</i> and <i>out strength</i> of node N_i .	The <i>in strength</i> and <i>out strength</i> related to incident and outgoing links of event N_i are added to indicate one facet of overall causal significance of event N_i in this particular network. The <i>strength</i> represents both the tendency of event N_i to be caused and to cause other events.
Betweenness centrality	$b(N_i) = \sum_{N_j, N_k \in N} \frac{P_{N_j N_k}(N_i)}{P_{N_j N_k}}$ Here $P_{N_j N_k}$ is the number of shortest paths connecting node j and k , and $P_{N_j N_k}(N_i)$ is the number of $P_{N_j N_k}$ that contains node i (Freeman, 1978).	The number of times of node N_i is located on the shortest path between two other nodes. While calculating the shortest path, edge weight is considered. The higher the edge weights, the more likely the path is to include one or more AND gates, indicating a longer “distance” between two nodes, further emphasizing the difficulties of causation via that path. Therefore, a shorter path is more likely to include one or more OR gates compared with AND gates.	A high level of <i>betweenness</i> indicates that event N_i appears in more causal pathways than other events. <i>Betweenness</i> is a measure of the overall causal importance of event N_i , where this measure is stretched over the whole network, taking both direct and indirect causal relationships with other events into account.
In closeness centrality	$C_{in}(N_i) = \frac{n-1}{\sum_{j=1}^{n-1} P_{N_j N_i}}$ Here $n-1$ is the number of reachable nodes from node j to i (Freeman, 1978).	The measure of the mean shortest distance from other nodes to node i . The shortest distance will be calculated using the same approach as described for calculating the shortest path in <i>betweenness</i> .	It measures the ease of event i to be caused by all its reachable (direct or indirect) events causally upstream.
Out closeness centrality	$C_{out}(N_i) = \frac{n-1}{\sum_{j=1}^{n-1} P_{N_i N_j}}$ Here $n-1$ is the number of reachable nodes from node i to j (Freeman, 1978).	The measure of the mean shortest distance from node i to other nodes.	It measures the ease of event i to cause all its reachable (direct or indirect) events causally downstream.

^aCausally upstream: The upper or earlier position in a sequence of events.

^bCausally downstream: The lower or later position in a sequence of events.

robustness curve. Thus, different risk treatment strategies can be compared on “*efficiency*.” The *efficiency* for using each metric can be derived from either the “area under the curve” or by identifying what is termed the *heel point*. Both are described below.

Referring to the example network depicted in Figure 6, we conduct two robustness analyses on it, with metrics *degree* and *out degree*, to demonstrate the concept. Comparing “area under the curve,” for example in Figure 6, provides a means of testing which metric is more efficient to perform the robustness analysis with for that particular network. A lower area under the curve for a particular metric compared with

another means that node removal is more “efficient” if the first metric is used as a guide rather than the second. This is because the metric’s curve with a lower area indicates that more nodes on average have been removed at each node removal step, compared with the other metric. From the lower figure within Figure 6, the *out degree* proves to be a more appropriate metric than *degree*, based on the values of the area under curve. This is further shown by the top-left tables in Figure 6 which capture the different sets of nodes being selected while performing these two analyses. This means that targeting events for risk treatment by considering only their number of consequences without also considering how

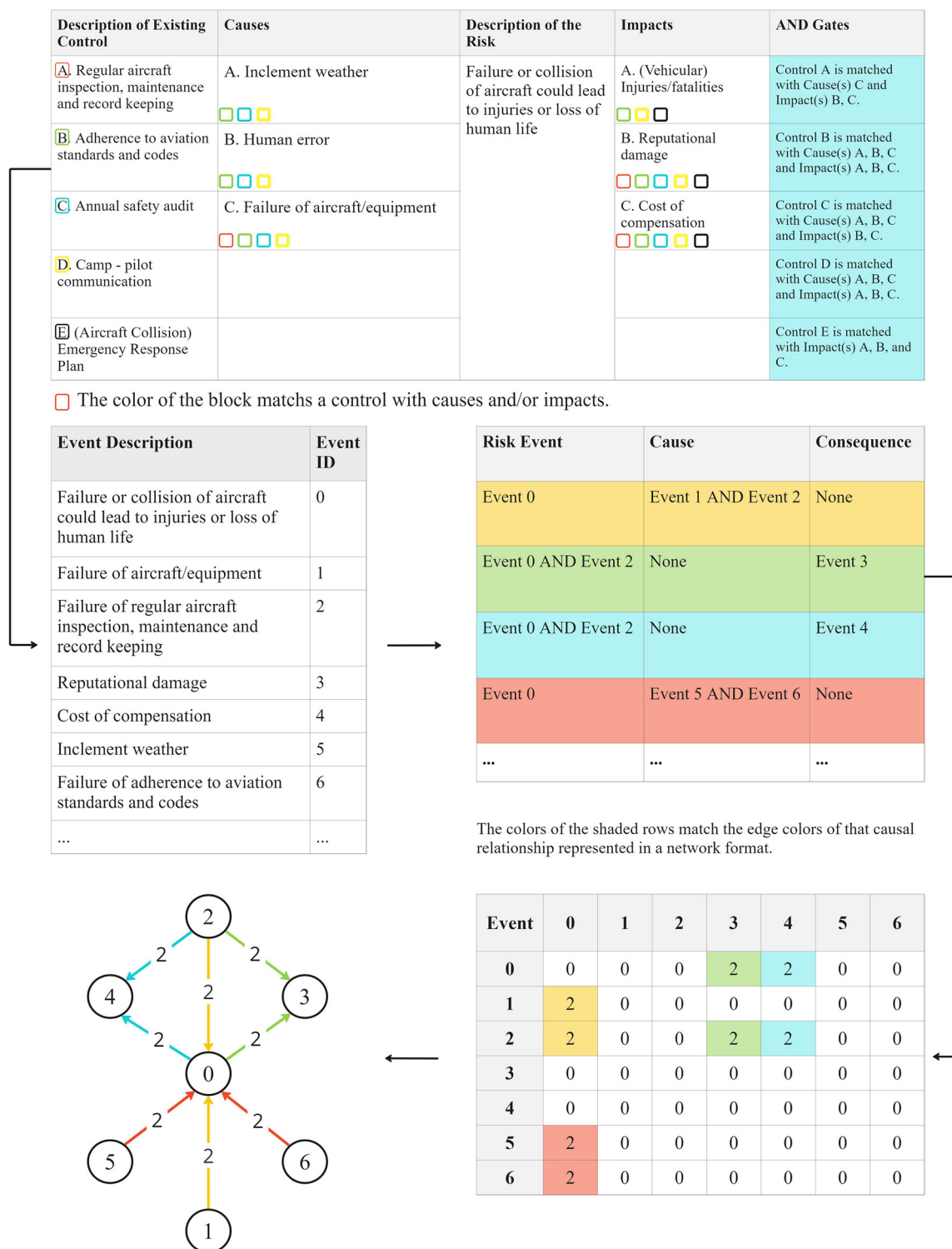


FIGURE 3 Conversion of risk register to formatted input data to adjacency matrix to network. The top table is the example risk register data. The middle-right table is the input CaNetA template, showing how the original risk data have been reorganized for analysis but not fundamentally changed.

many causes they have indicates a more efficient risk treatment in this case. In these ways, robustness analysis can be used to indicate a bespoke risk treatment strategy for the particular causal network under investigation.

Another way to measure the *efficiency* is the *heel point*. The *heel point* of a robustness curve is defined as the ratio of nodes that have to be removed to eliminate all the nodes in the network compared with the initial number of nodes in the network. This gives a percentage measure that is an indication of

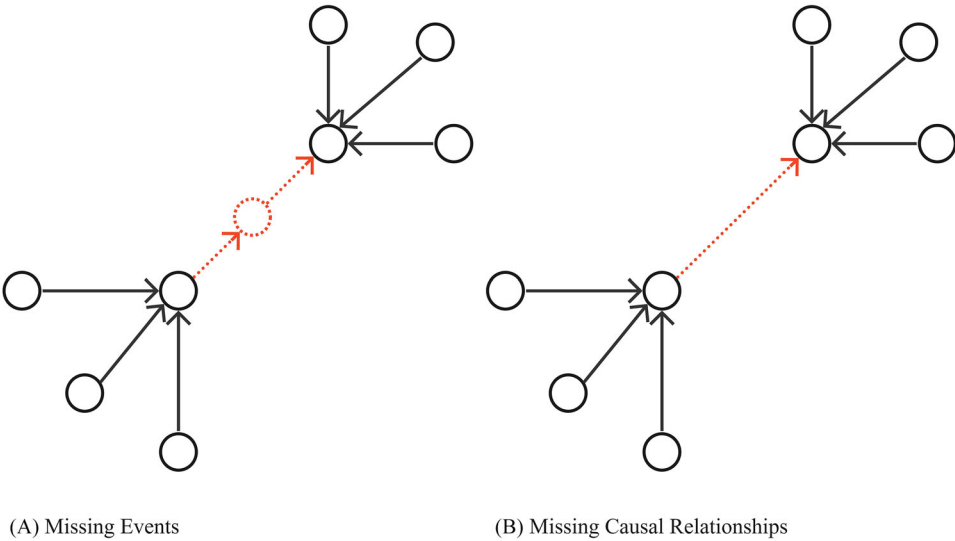


FIGURE 4 Example of how two separate networks derived from a risk scenario may be the result of poor quality input data. For instance, missing events (red dotted circle) or causal relationships (red dotted line).

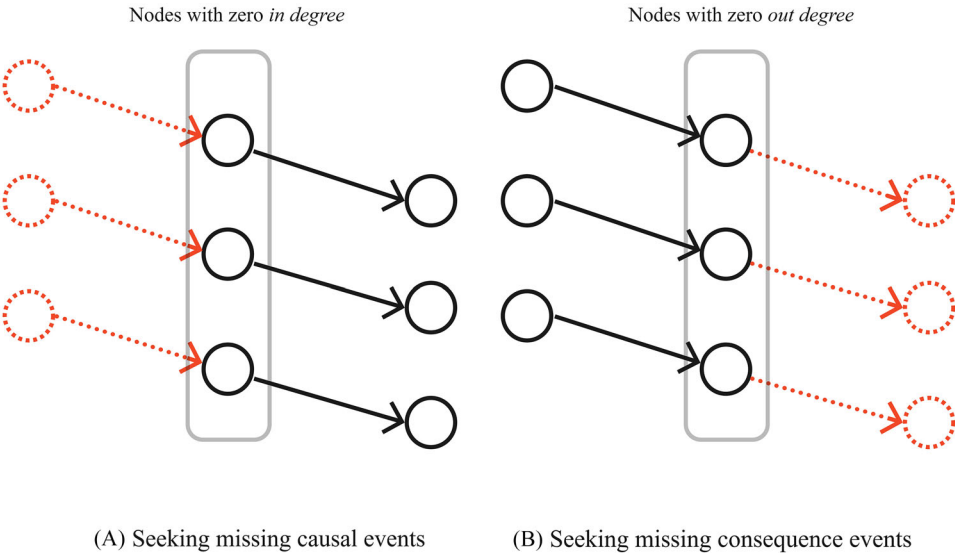


FIGURE 5 Example of using topology metrics to improve risk assessment quality—inspecting the network to check if the risk assessment scope has been achieved.

node removal efficiency, and a proxy measure, albeit a simplification of risk treatment strategy effectiveness. In this study, this kind of strategy is also shown in Figure 6. It shows that *out degree* has a lower value of *heel point* (25%) as compared with *degree* (33%), which suggests that follow *out degree* for node removal is more efficient.

We further propose a strategy named “Total Combined,” which combines individual metrics, based on the following process:

1. Calculate the descending order for all the node values based on the topology metrics of interest (for example using the 9 metrics described in Table 2).

2. For all of the topology metrics, pick the node that has the highest value for each metric being compared and remove the single node that, when eliminated, results in the greatest reduction in the number of other nodes within the network.
3. Iterating steps 1 and 2 until there are no more nodes within the network.

Figure A1 presents a detailed flow chart that illustrates the process contributing to “Total Combined.” This kind of strategy can be extended to an approach that targets two different metrics displaying the highest partial gradients among all metrics, and then combines them. A partial gradient

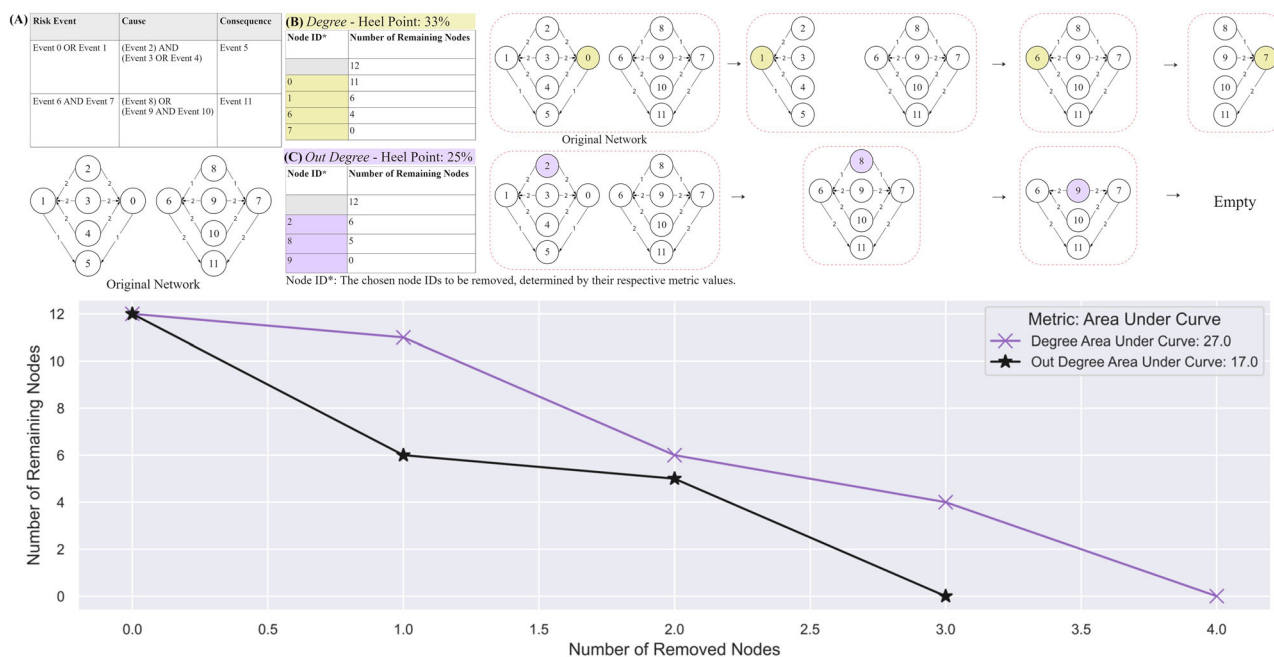


FIGURE 6 The robustness analysis on an example network, illustrating how to calculate which metric is most efficient.

means the gradient at a specific segment of a robustness curve. A metric robustness analysis with a higher partial gradient indicates that node removal can be accelerated by following that analysis, with that partial period. This concept will be further elaborated with an illustrative example in Section 3.3.3.

Typically, risk treatment strategies are based on prioritizing the highest risk first in descending order to lowest risk. Various measures of risk can be used for this, or purely consequence or likelihood in some circumstances. We introduce the term *consistency* to evaluate how well the data from the risk register align with the results from robustness analyses across different metrics. A higher *consistency* indicates that more risk events (events listed in the column named “risk event”) are captured within the robustness analysis. Therefore, the measure of *consistency* quantifies the extent to which risk events are captured within the robustness analysis, irrespective of their actual risk score. We are interested in examining the correlation between the *efficiency* of the risk treatment strategies derived by robustness analysis, and the *consistency* with the risk information stored in the risk register. Consider Figure 6 as an example to derive the *consistency* of two robustness analyses. The upper-left table Figure 6(a) indicates that event numbers 0, 1, 6, and 7 are risk events, which are documented in a risk register. Figure 6(b) and (c) reveals that *degree* robustness analysis has a *consistency* of 100% (all the events selected are risk events), while *out degree* robustness analyze has a *consistency* of 0% (none of the events selected are risk events).

3 | ILLUSTRATIVE EXAMPLE: MINE PROJECT FEASIBILITY RISK ASSESSMENT

This section illustrates an application of CaNeTA to a mine project feasibility study to demonstrate its essential features.

3.1 | Description and data collection

This study involves the analysis of a Workplace Risk Assessment and Control (WRAC) developed for a mine project feasibility assessment, captured in a risk register. The project involves constructing production and transportation facilities to deliver metal concentrate. The risk register includes the overall phases of the project, from the issuing of permits to operation, through to post-closure. The data are deidentified.

3.2 | Data analysis

The project risk register is the input dataset for this analysis. The data extracted from the risk register first went through the preprocessing step. Figure 3 shows part of the risk register being transferred into the causal network. For brevity, the entire risk register cannot be shown.

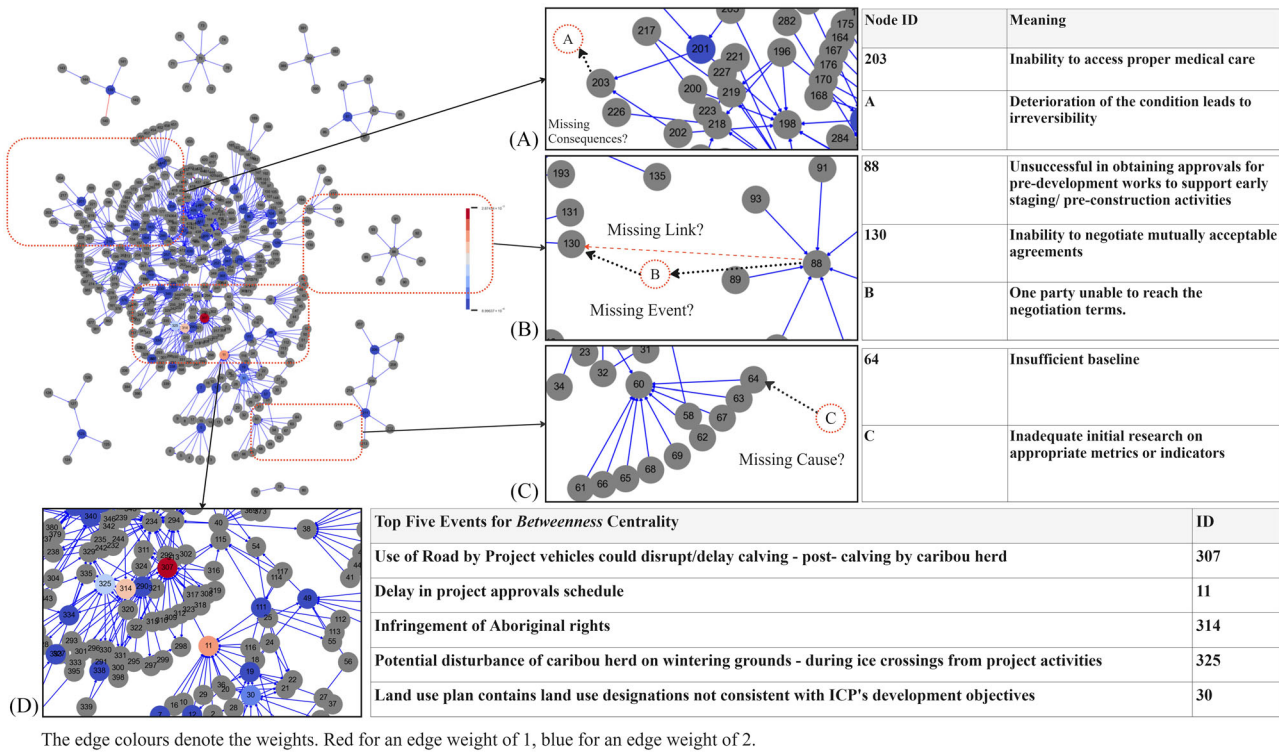


FIGURE 7 Causal network developed from the example risk register: (a) There might be a missing consequence A, linking to event no. 203. (b) There might be a missing causal relationship between event 88 and event 130, or a missing event B between these two events. (c) There might exist a missing cause C to event no. 64. (d) Representative examples of events with high betweenness. The graph is produced by NetworkX (Hagberg et al., 2008).

3.3 | Results

3.3.1 | Visualization

Applying CaNeTA to the data described in Section 3.1 results in the causal network illustrated in Figure 7. The network contains 473 nodes representing 473 unique events, showing their logical relationships, allowing network visualization. The events are recorded in Table B3. Several smaller subgraphs surround the largest central subgraph, which suggests that the risk register may be missing some events and/or causal links. We take this view because in risk assessment, where you can have a well-defined scope focused on a particular system, one would expect that events are clustered and interrelated, ideally forming a network with no disconnected subgraph. Examples of where analysts could look for these missing pieces are indicated on the top-right side of Figure 7.

3.3.2 | Individual metrics

Figure 8 includes two examples of the colormaps that represent both the *out-degree* and the *in-degree* distributions of nodes, showing that node 162 (Failure of worker training) and node 249 (Failure of mineral waste management plan) have the highest and second highest *out-degree* values among all nodes, followed by nodes 421 (Failure of collision risk management system—emergency response), 422

(Failure of collision risk management system—contingency procedures), and 160 (Failure of road management plan). The development of the colormap of the causal network for the risk register gives a direct visualization of the nodes based on their metric values, which can be used to prioritize the events that require attention.

As emphasized in Section 2.3, the usage of topology metrics can indicate the causal importance of individual events, having explicitly taken into account their causal context. Figure 7(a) exemplifies the fact that node 203: “Inability to access proper medical care” is a *final consequence*, because it has the *out-degree* value of 0. In contrast, Figure 7(c) shows node 64: “Insufficient baseline” has no causes and thus indicates a *root cause* within the scope of that particular risk assessment. These examples signal that there might be other consequences linked to node 203 or other causes leading to node 64. However, as the network becomes more complex, investigation of *final consequences* and *root causes* using this kind of analysis becomes cumbersome. Figure 8 records “root causes” and “final consequences” in the context of the case study, which are extracted using the metrics *in degree* and *out degree*.

Similarly, other metrics provide information on which events are more likely to be “hub” or “critical” events, for example, events with high *strength* or *betweenness*. Figure 7(d) includes the top five events with significant values of *betweenness*. The event, “Use of Road by Project vehicles could disrupt/delay calving—post-calving by cari-

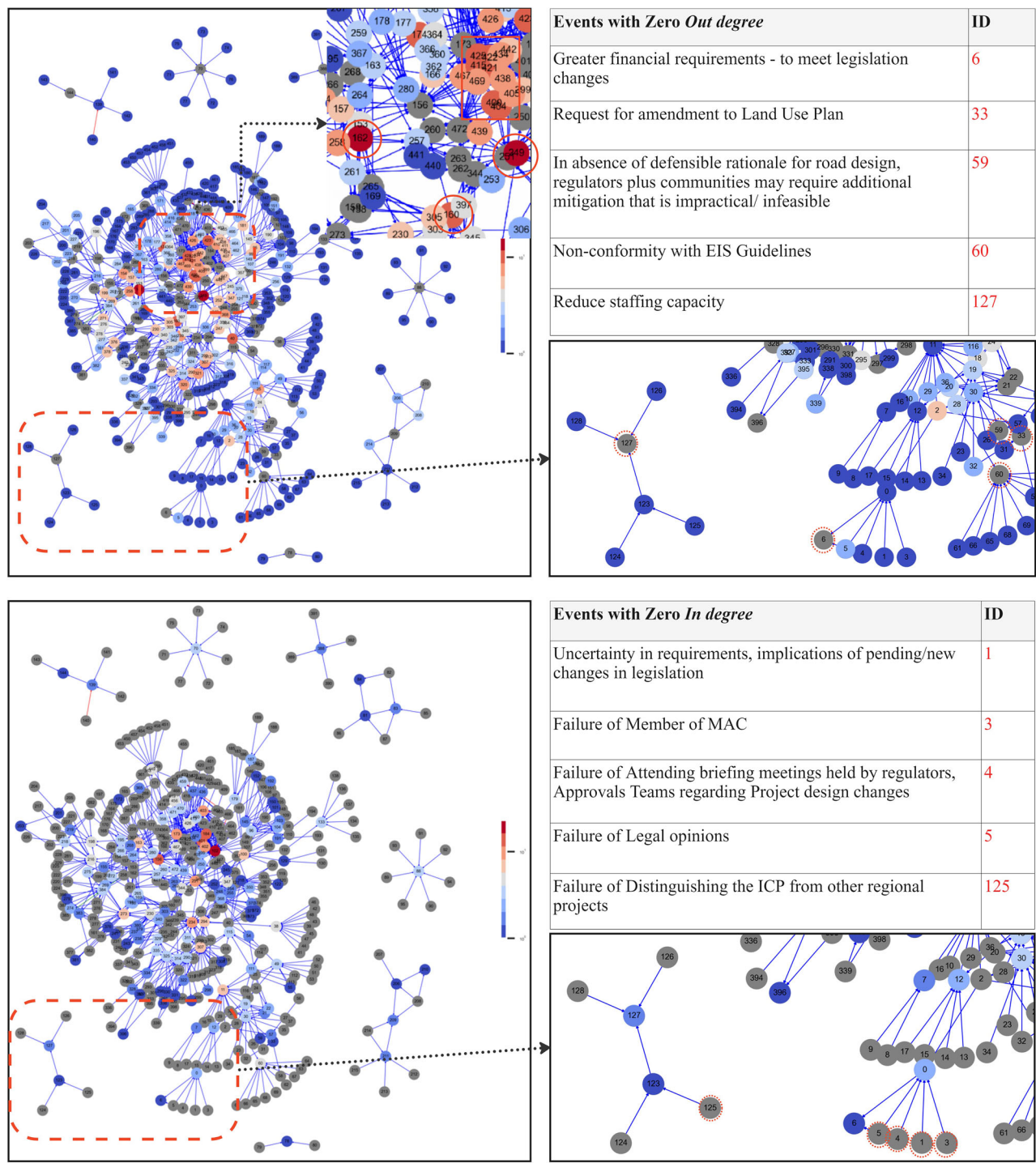


FIGURE 8 In-degree and out-degree colormaps developed from the example risk register, with representative examples of events with zero out-degree or zero in-degree (graph produced using NetworkX; Hagberg et al., 2008).

bou herd” has the highest *betweenness* value. These metrics indicate what kinds of events drive the whole causal scenario.

The example risk register records likelihood and consequence data for each risk event, and an overall risk level for a risk event is also determined by using a risk matrix. However, the risk level used, being “high,” “medium,” and “low,”

only enables risks to be differentiated into these three categories. Prioritization beyond this level of resolution is not possible based on risk levels alone. There are 11 risk events categorized as “high,” 16 as “medium,” and 24 as “low.” However, risk events can meaningfully be prioritized, within each risk level, by using topology metrics. Figure 9 shows the differentiation of these risks with the same risk level,

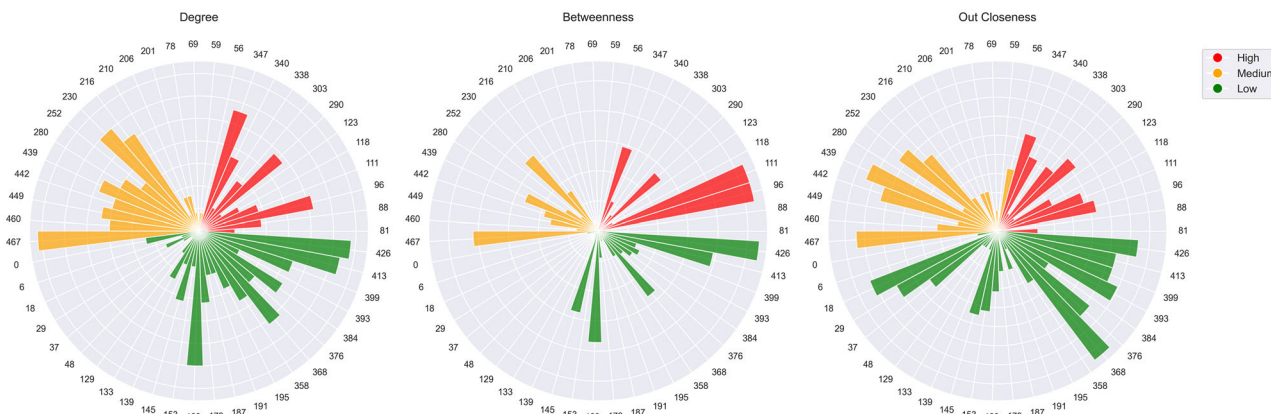


FIGURE 9 Three examples (degree, betweenness, and out closeness) of the metric histogram. The radial axis represents all the risk events with a risk level (high, medium, and low) scored from the risk register. The histogram bars represent the metric values of the risk events.

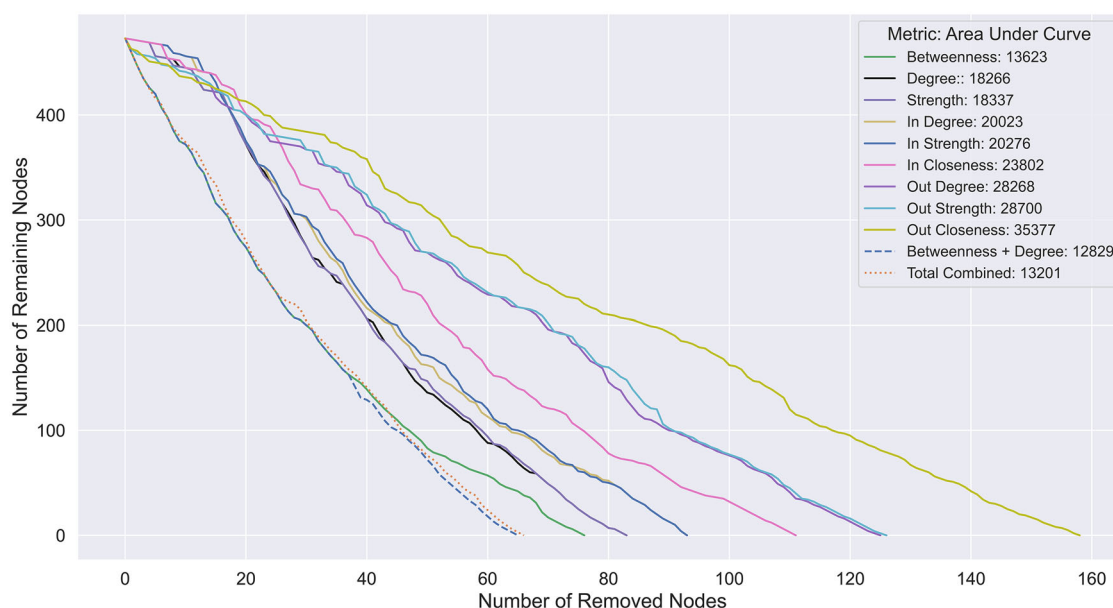


FIGURE 10 All metrics robustness analysis.

based on three example metrics: *degree*, *betweenness*, and *out closeness*. This information can be used to make a meaningful distinction between risk events, and to prioritize between these events, after accounting for the differences in their risk score values.

3.3.3 | Robustness analysis

The node elimination trends from the robustness analysis are shown in Figure 10. The *betweenness centrality* behaves as the most effective individual metric for prioritization of node removal. Though *betweenness* performs an optimal outcome, after 40 nodes have been removed the gradient of the curve is no longer the highest, compared with the other metrics' robustness curves. This implies that switching from *betweenness* to other metrics (*degree*

in this case) from the 40th removed node could create a better result overall for the robustness analysis. Therefore, a combination metric, "*Betweenness plus degree*" is introduced. For the first 40 nodes, we choose the nodes with the highest value of *betweenness* to remove. After 40 nodes, we switch to the nodes with the highest *degree* value.

We further calculated the optimal metric with the highest *efficiency* in each step of node removal ("Total combined," illustrated with the dotted line), in Figure 10. However, the results show that the highest *efficiency* in each step does not produce the most optimal strategy. From Table B1, we deduce that the combination of *betweenness* and *degree* performs most efficiently during the robustness analysis, as seen in the *heel point* and area under the curves. The three metrics, which have the best performance from the robustness analysis with the results, are summarized in Table B2.

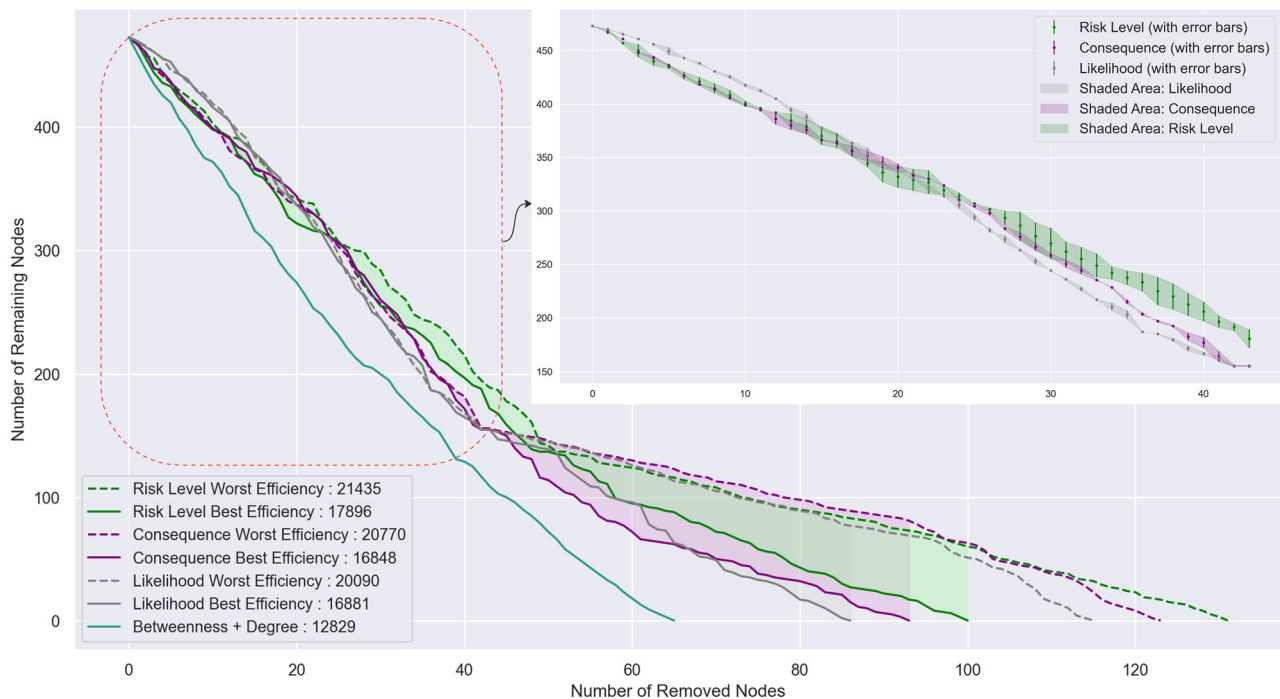


FIGURE 11 Robustness analyses comparison: betweenness plus degree vs. risk level vs. consequence vs. likelihood.

We take advantage of the network robustness, to compare various node removal strategies based on the risk information: risk level, likelihood, and consequence. There are a total of 51 risk events in the input dataset with an attached risk level categorized as “high,” “medium,” or “low.” Forty-two of these risk events had likelihood and consequence data, ranging from 1 to 5, 1 being the lowest and 5 being the highest. For instance, considering “risk level” for robustness analysis:

1. For risk events with risk level ranking, randomly pick one from these events with a “high” ranking and remove that event from the network.
2. Iterate until there is no risk event with a “high” ranking.
3. Repeat steps 1 and 2, first changing the ranking from “high” to “medium” and then repeat steps 1 and 2 with the ranking of “low” until all 51 risk events are removed.
4. Randomly remove remaining events until all the events have been removed.

Undertaking the same process using likelihood and consequence data yields three different robustness analysis results. Given the fact that randomness does cause uncertainty, we have conducted 500 simulations for each type of risk information and captured the robustness curves with the highest and lowest *efficiency*. These results are presented in Figure 11 and Table B1 “8Risk information” column. Among the analyses, the likelihood and consequence share similar *efficiency*, while the risk level provides the least *efficiency*. These results are not as efficient a treatment strategy as the topological robustness analyses using either the “Total combined”

or “*betweenness plus degree*.” In other words, by choosing nodes in the order of the “high,” “medium,” and “low,” risk score does not provide the best results for robustness analysis. To enable a more precise comparison between the robustness results, we have excluded the random node removal after node 42, as shown in the inset figure of Figure 11. The error bars indicate the uncertainty in the data, calculated as half the difference between the max (curve with the worst performance) and min (curve with the best performance). Additionally, it emerged that the level of uncertainty from the randomness inherent in the risk level robustness analysis is significantly higher than that in the likelihood and consequence analyses.

Figure 12(a) shows the risk level of events in their order of removal during robustness analysis, following three different metrics. First, it is clear that for some risk treatment strategies, more risk events appear in the subset of nodes that are selected for removal. This means, that the treatment strategy has a higher *consistency*. Other metrics chosen for robustness analysis have a lower *consistency*: see the differences between the three analyses in Figure 12(a). Second, regardless of the metric used in robustness analysis, the risk level of the removal of the event does not correlate with the order these events are removed. Furthermore, the relationship between the metric *consistency* and the *efficiency* is illustrated in Figure 12(b). Each dot point represents different risk treatment strategies from different metric robustness analysis results. Both plots show that higher levels of *consistency* correlate with increased *efficiency* in metric robustness analysis, for the risk register being studied.

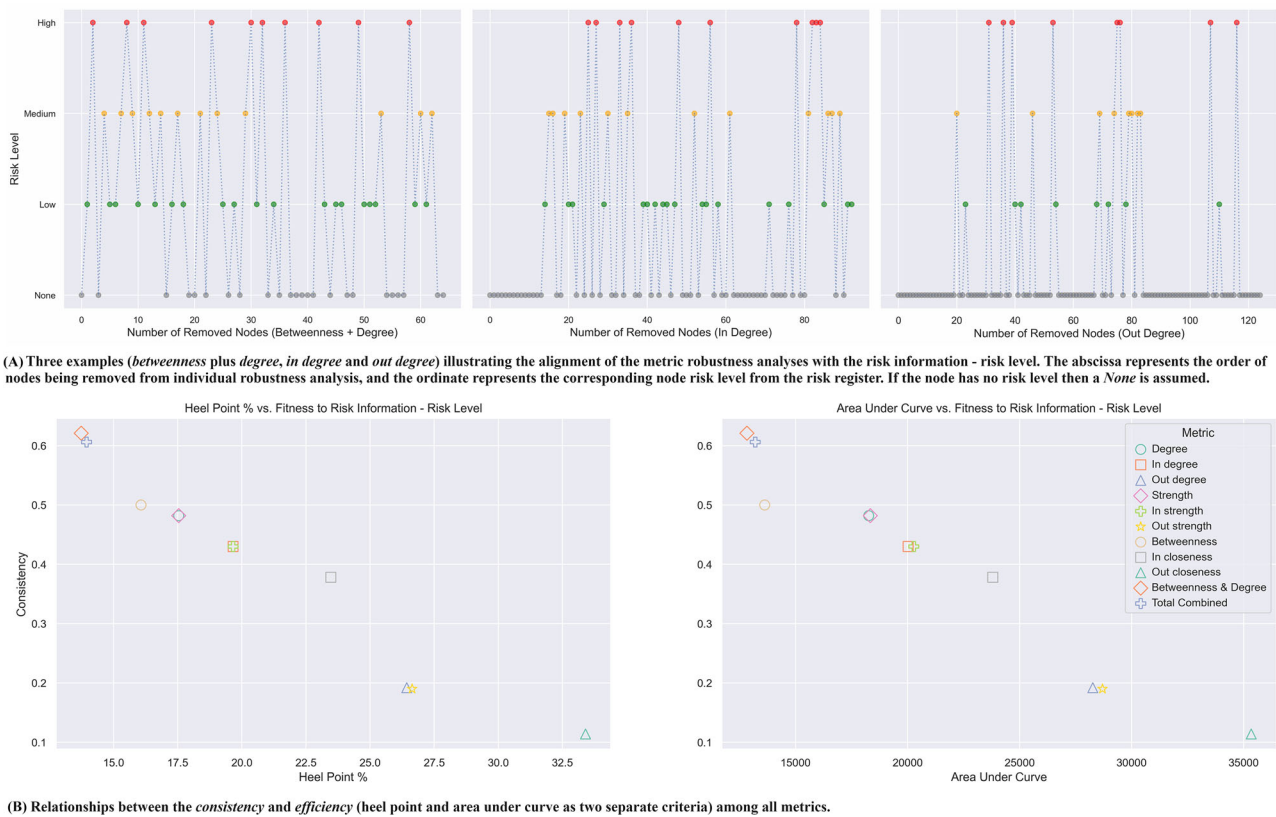


FIGURE 12 Combine risk information–risk level with the robustness analysis.

3.3.4 | Key results summary

The following are several key insights extracted from the results:

- There are events important to the progression of causal scenarios that topology metrics can identify.
- Risk events with identical risk scores can have very different topology metrics scores. Using topology analysis helps improve the precision of risk prioritization.
- According to the robustness curves illustrated in Figure 10 and Table B1, focusing management on 76 events of 473 will significantly reduce the overall project risk.
- Using risk scores for risk treatment is less efficient than using the “*betweenness plus degree*” in robustness analysis and may not remove all the events in the causal scenario.
- There appears to be no correlation between the risk level and the sequence of node removal as determined by any metric robustness analysis.
- Metric robustness analysis with a higher *consistency* also exhibits a higher *efficiency* of risk treatment.

These conclusions will be elaborated in more detail in Section 4.

4 | DISCUSSION

In this study, we have applied network topology metrics to characterize the causal context of risk events. The calculation of network metrics and the interpretation of these metrics provide suggestions for risk prioritization and treatment decision-making. Compared with traditional methodologies, CaNeTA enables analysts to characterize the causal context of their system or project risks. This demonstrates how individual risks are structured and connected and helps identify critical events. Table 3 highlights how CaNeTA contributes to improving current topology analysis approaches as mentioned in Section 1.

Data-driven analyses use historical data to obtain networks and graphs. These networks do not focus on the relationships between each risk factor. They build the scenario based on past experiences, group similar risk events into one category, and link these categories to form the network. This approach can miss insights created by a group of analysts seeking to understand a specific type of scenario for a specific project. Real causal complexities present in the detail of specific risk assessments, useful or even required for successful decision-making, can be obscured by the summarizing of some data-driven methods.

In contrast to dynamic risk analyses, our work has a different scope by providing identification of event criticality

TABLE 3 How CaNeTA contributes to past approaches and supports risk management decision-making.

Contributions	What CaNeTA offers
Articulation of the network topology analysis so that it is repeatable for multiple risk management problems	The translation of risk expressed in different data sources, from diverse disciplines, into one format to enable integration
Definition of the set of concepts used to extract and combine data to form the networks	Confer the meaning of individual elements (nodes) and the relationships (edges) between these elements, generalize the meaning of “event” and “cause,” and provide clarity on AND/OR gate structures. Any kind of risk can be connected in the causal network from all risk categories (safety, health, environment, finance, etc.).
Guidance on how to justify the selection of which network metrics to use, for particular patterns or questions	Define the interpretation of a common set of metrics applied under a risk management context that have been described previously in Section 2.3 and will be further elaborated in Section 4.2.
Interpretation of how metric results can be used for characterizing the causal context, thus informing risk management decision-making	Network visualization, calculation of network metrics, and network robustness results enhance comprehension of causal scenarios within risk data, characterize the causal context, and support risk management decisions. For example, the prioritization between risk events and the usage of multiple topology metrics can potentially add a relevant and complementary approach to risk analysis for prioritizing which risks are causally important, and how the topology can be used to support the development of risk treatment plans.

and strategies for risk treatment, without predicting or updating probabilities of event occurrences. Nonetheless, the way that dynamic risk analyses integrate probability data, particularly those using Bayesian networks (Adedigba et al., 2016; Chen et al., 2021; Khakzad et al., 2012, 2014, 2017; Liang et al., 2017; Rathnayaka et al., 2011a, 2011b; Shi et al., 2024), offers valuable insights that can be adopted. For instance, likelihood data for events within a risk register can be converted into relative probability data and assigned as weights to the nodes. For those events/nodes without any likelihood data, Bayesian analysis can be applied to derive their probability data. The node weights then serve as another metric for comparison during the robustness analysis.

Apart from this work, CaNeTA has also been applied in fire safety (Lin et al., 2023), to offer risk analysts insights into the overall system performance, and provide suggestions on the criticality of fire safety components.

4.1 | Causal network

Events describe change as a transition between states. In this paper, we have shown that sequences of events form a causal scenario. These events, regardless of their data sources, are transferred into standard language about risk, causes, and consequences and then converted into a causal network. Events are precisely defined, categorized, and embedded within logical gates depending on their causal relationships.

4.2 | Individual metrics

We have explored the deeper meanings of some established network metrics, by examining their importance and impli-

cations for people making risk management decisions. In Table 4, we generalize *interpretations* for some of the more common and important metrics. These interpretations can aid in selecting network metrics for CaNeTA, whereby analysts could choose metrics from this list to match the particular questions they have for their risk data.

The value of the insights from any individual metric lies in its ability to reveal what is typically unforeseen. One of the supporting observations from the case study provides an example. The significance of caribou to the success of the project is surprising as it was not highlighted as such in the original risk register. In Figure 7, the event “Use of Road by Project vehicles could disrupt/delay calving—post—calving by caribou herd” has the highest *betweenness* value. The company that provided the case study commented that caribou-related issues like this were arising as significant in their discussions with indigenous people. However, this event did not have a significant risk rating in the original risk assessment the company performed (as shown in Figure 9, this Caribou Event 307 is absent). This is initial evidence that interrogating the topology of a causal network can indicate events of causal importance, some of which are critical to project success, in a way that augments traditional risk management practice. In short, if risk analysts only use “risk” as the measure, key risks can be missed. Interrogating the topology of causal networks is one way to identify them, and it enriches the understanding of the significance of these events.

4.3 | Robustness analysis

The process of the robustness analysis has been illustrated with an example in Section 3.3.3 and suggests a risk treatment strategy that is efficient in terms of breaking apart the risk

TABLE 4 Topology metrics meanings interpreted in the field of risk management.

Metric	Meaning and use in risk management practice
Degree	Risk analysts could use this metric to find the more causally well-connected events, regardless of the logical relations between events, focusing more on the ability to be caused and directly cause other events.
In degree	This metric gives information on the tendency of events to be caused by other events.
Out degree	This metric gives information on the ability of events to cause other events to happen.
Strength	The edge weights for each link of an event indicate whether that edge is within an AND or an OR gate. For example, if two events have the same <i>degree</i> , the <i>strength</i> value is considered higher if more events are within an OR gate. Because <i>strength</i> combines the tendency of an event to be caused, and the event's ability to cause other events, <i>strength</i> could be a candidate proxy measure of <i>risk</i> associated with that event based on topology alone. This is a relative measure only, comparable with the other events in the particular network of interest.
In strength	Risk analysts can derive more insight into direct causation than gained from <i>in degree</i> . For example, consider two events with the same number of incoming edges, one with all these edges within an AND gate, and the other having these edges within an OR gate. The latter event is considered more likely to occur, as multiple causes lead to that event independently. The latter event would also have a higher value for <i>in strength</i> than the former. <i>In strength</i> may be a proxy value for a kind of relative likelihood, compared with other events in the network of interest.
Out strength	This metric indicates the ability of events to cause other events, including details of their logical relationships. In addition, events will exhibit a lower value of <i>out strength</i> in comparison with their <i>out degree</i> when they are involved in more AND gates.
In closeness	This metric gives risk analysts a global understanding of each event, describing how tightly the event is connected to all other upstream events that might cause it. A higher value of <i>in closeness</i> might indicate that this event can be caused by other events through multiple paths. The calculation of this metric considers the logical relations between events. Consider two paths from other causal events to a particular risk event; the first path has more edges with OR gates than AND gates when compared with the second path. The total distance for the first path is considered to be shorter. When two events have a similar number of paths pointing from other events, a higher number of OR gates on those paths will yield a higher value of <i>in closeness</i> , and reflects that it has a higher tendency of being caused across the whole network, considering both direct and indirect causal connections.
Out closeness	This metric gives risk analysts a global understanding of each event, describing how tightly the event is connected to all other downstream reachable events that might be caused by it. A higher value of <i>out closeness</i> might indicate that this event can cause other events through multiple paths by sequences in an overall sense. Similar to <i>in closeness</i> , consider two paths from one risk event to two other consequences. The first path has more edges within OR gates than AND gates when compared with the second path. In that case, the total distance is shorter. When two events have a similar number of paths pointing to other events, a higher number of paths with OR gates will yield a higher value of <i>out closeness</i> , and reflects that it has a high tendency to cause other events to occur, considering both direct and indirect causal connections.
Betweenness	This metric determines how often one event is located on the shortest path between two other events, with the association defined as the edges connecting these two events. The <i>betweenness</i> represents the number of causal pathways that flow through that event, or that that event belongs to. That is, because a causal network represents the overlap of many potential simpler causal pathways that may occur, a high <i>betweenness</i> score for an event shows that the event is involved in more potential causal pathways than others. This indicates a global likelihood—if an event exists in more potential scenarios in general, it is more causally important and contributes to more undesired scenarios progressing. This is distinct from proxy likelihood measures derived from <i>in strength</i> or <i>in degree</i> and brings an extra dimension to the analysis. This in turn may mean that high <i>betweenness</i> events should be considered early on in risk treatment, as key places to apply risk controls.

landscape represented by the network. The interpretation of this analysis is that the node removal process can act as a model of a risk treatment process. Table 5 concludes such interpretation results.

In summary, traditional risk management practice does not consider the topological information bound up in the causal context, particularly during the risk prioritization which directly influences the *efficiency* of a risk treatment strategy. In this specific risk register, events listed in the “risk event” column that the analysts indicated, are also important topologically. By focusing on preventing these events, regardless of their attached risk scores, risk analysts can conduct a more effective risk treatment strategy. However, different risk registers may not follow this pattern. Conducting robustness analysis among multiple risk registers can help character-

ize different networks based on the *efficiency* of their risk treatment strategies. This approach offers a straightforward comparison between risk registers, highlighting the similarities and differences in their management of risks within their specific causal context. Additionally, risk data that contain causal relationships between elements can also be analyzed and compared for a deeper understanding. Future work will elicit this.

5 | CONCLUSIONS

This study raises the issue of the lack of integration of causal context in traditional risk assessment practice. CaNeTA helps to address this issue and can be used to characterize the

TABLE 5 Robustness analysis interpretation results.

Robustness analysis results	Interpretation
All metrics robustness analysis from Figure 10.	This result illustrates the performance outcomes of robustness analyses, conducted using different topology metrics. Efficiencies of risk treatment strategies can be derived from these metrics. These strategies can be compared by the “area under the curve” and <i>heel point</i> . An example of a risk treatment strategy can be: Following the descending order of the <i>betweenness centrality</i> to identify how to prioritize events. Of a total of 473 events, only 76 events are required to be prevented to greatly reduce the overall risk of a scenario. As such, prioritizing the control of the 76 events may be just as effective as attempting to implement controls for the entire network.
Robustness analyses comparison—“ <i>betweenness</i> plus <i>degree</i> ” vs. risk level vs. consequence vs. likelihood from Figure 11.	We can deduce that by choosing events to deal with, following the order of risk level (from high to low), likelihood (from 5 to 1), or consequence (from 5 to 1) will reduce the immediate risks but might not address all the events in the causal scenario.
The alignment of the metric robustness analyses with the risk information from Figure 12(a).	Combined with Table B1, these two results suggest that there is no correlation between the risk level and the order of node removal according to robustness analysis. This suggests that the order of treating risks according to the risk level (high to low) does not align well with the order of treating risks following topology metric robustness analysis. It should be noted that these two treatment strategies (risk level and metric robustness analysis) have different objectives. The former addresses events individually, with no consideration of the cumulative impacts of causal interrelationships between multiple risk events. Whereas the latter addresses the causal interrelationships without consideration of the impacts of the individual events. Therefore, there is value in bringing these two approaches together; however, a full discussion of this is outside the scope of this paper.
Relationships between the <i>consistency</i> and <i>efficiency</i> among all metrics from Figure 12(b).	It appears to show a linear correlation between <i>efficiency</i> and <i>consistency</i> for this particular case study. A higher <i>consistency</i> with the risk register correlates positively with an increased <i>efficiency</i> in risk treatment according to robustness analysis. This means that, in this particular risk register, focusing on treating events under the “risk event” column is likely to be an efficient strategy regardless of their risk scores.
The combination of <i>betweenness</i> and <i>degree</i> performs most efficiently during the robustness analysis, from Table B1.	It suggests the potential existence of other topology metrics beyond the scope of this study, which might be able to generate better robustness results compared with analyzing the existing metrics. In addition, modeling risk treatment through choosing the metric that eliminates the highest number of events in each step might not generate a globally optimal prioritized risk treatment strategy, because we have only considered nine metrics here.

causal context and provide insights for risk management decision-making. Compared with previous network topology analysis methodologies, CaNeTA contributes to filling gaps in understanding the *terminology* and *interpretation of network metrics* by (1) articulating network topology analysis in a repeatable way for diverse risk management problems, (2) defining a set of concepts to extract and combine data to form networks, (3) interpreting the meanings of network metrics with their values, and (4) providing guidance on the selection of network metrics.

This approach is illustrated through its application to a mining example. It is shown that CaNeTA can combine different kinds of risk information from diverse data sources to provide richer insights, enabling a deeper understanding of the causal scenario. The causal linkages among multiple events can be integrated into one causal network, and colormaps provide visualizations based on the locations of the events within the network. The topology contains informa-

tion, extractable through the calculation of various network metrics (Barabási & Albert, 1999; Boccaletti et al., 2006; Newman, 2000), which informs:

1. The role that individual risks play in their causal context, influencing how their importance is determined, beyond considering the risk score alone.
2. The characterization of the causal context as a whole, enabling richer decision-making about the management of all organizational risks.

Characterizing the whole network provides a holistic analysis of risks. Risk events with similar risk scores can have very different network metric scores, which provides an additional level of information for decision-makers to prioritize between events. Similarly, robustness analysis provides an alternate and transparent pathway for the prioritization of risks and can suggest an idealized optimal order within which

to treat risks. Another learning from the current work is that the structure of a causal network can be sensitive to changes in its nodes and edges which directly affects topology analysis. This insight helps to connect two sources of uncertainty in risk management processes and network topology together. Consider two fundamental aspects of risk management: the number of risks a risk assessment should include (its scope) and the level of details required for describing a risk, its causes, and consequences (event granularity). These two aspects introduce additional uncertainty into a risk management process. By studying the phenomena when there are modifications in network structure, possible guidance of decision-making on the scope and granularity could be provided.

There is more that can be done to improve CaNeTA's utility. Future work will entail:

- Investigating uncertainty and/or subjectivity associated with CaNeTA as a method and comparing it to the uncertainty and/or subjectivity present in traditional risk assessment processes.
- A directed graph cannot distinguish events within AND gates effectively. For instance, if event A, with the occurrence of *either* event B or event C, leads to event D, this relationship may not be clearly discernible. Addressing this limitation will be important to improve the clarity of the analysis.
- The outcome from robustness analysis limits itself to considering *efficiency* of node removal only. This highlights that the way we are performing the robustness analysis is only a model, an approximation of a risk treatment strategy. Incorporation of node weights into robustness analysis, which represent severity, likelihood, and/or cost of treatment, will enhance our approach, connecting it more deeply to real risk treatment practice.
- There exist opportunities for linking CaNeTA with dynamic risk analysis models to illustrate the effectiveness of risk assessment strategies. For example, by taking the initial occurrence probabilities of network outermost layer events and updating the occurrence probabilities for all events within the network, a comparison between the likelihood of risk events in the input risk assessment and the occurrence probabilities of risk events after being updated can be delivered. Events with no occurrence information, for example, the consequence events, can also be updated (Khakzad et al., 2013a).

In this paper, we have shown that CaNeTA is a transparent and formal way of quantitatively incorporating an analysis of the causal context into risk management practice. Although there remain some limitations to this approach, it can complement and augment existing risk management activities.

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APPENDIX A

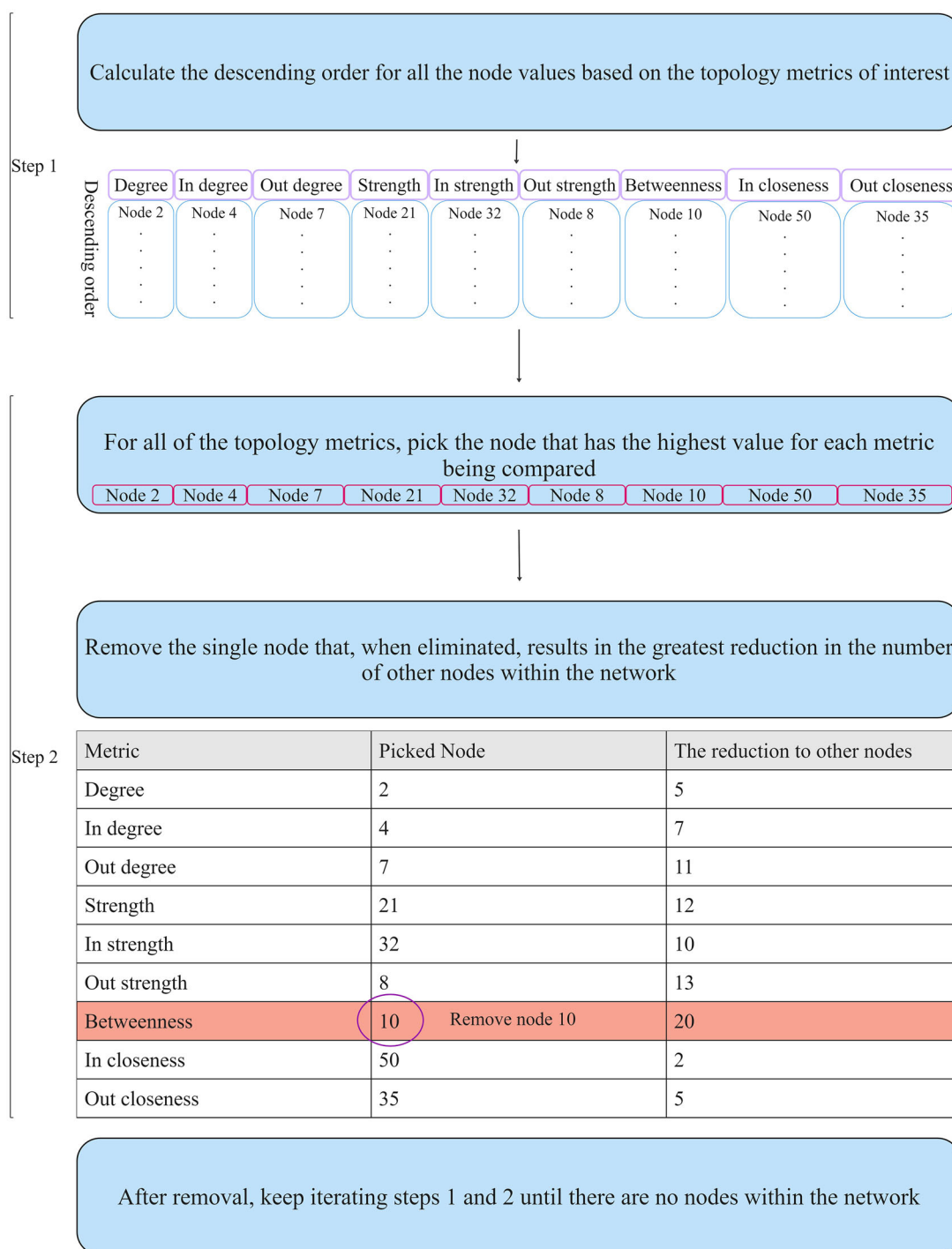


FIGURE A1 Metric 'total combined' process flow chart.

APPENDIX B

TABLE B1 Numerical results interpreted from the robustness analysis.

	Heel point %	Area under curve	Consistency (%)
Individual metric			
Degree	17.54	18,266	48.2
In degree	19.66	20,023	43.0
Out degree	26.43	28,268	19.2
Strength	17.54	18,337	48.2
In strength	19.66	20,276	43.0
Out strength	26.64	28,700	19.0
Betweenness	16.07	13,623	50.0
In closeness	23.47	23,802	37.8
Out closeness	33.40	35,337	11.4
Combined metrics			
Betweenness plus degree	13.74	12,829	62.1
Total combined	13.95	13,201	60.6
Risk information			
Likelihood	17.97	16,881	
Consequence	19.45	16,848	
Risk level	20.93	17,896	

TABLE B2 Results of the top three metrics with the best performance from the robustness analysis show the node IDs removed at each node removal step.

Node removal step	Betweenness and degree	Total combined	Betweenness
1	307	307	307
2	426	426	426
3	96	96	96
4	325	162	325
5	467	325	467
6	413	467	413
7	163	413	163
8	230	347	230
9	347	163	347
10	439	249	439
...	...	...	...

TABLE B3 Node IDs with their meanings.

Event description	ID
New regulatory—legislative requirements	0
Uncertainty in requirements, implications of pending/new changes in legislation	1
Failure of communication with regulators	2
Failure of being the member of Mining Association	3
Failure of attending briefing meetings held by regulators, and approvals teams regarding project design changes	4
Failure of Legal opinions	5
Greater financial requirements—to meet legislation changes	6
Late changes to project design	7
Project change that results in additional as well as other regulatory requirements that were not identified earlier in the permitting phase	8
Failure of regulatory review, update permitting strategy for all project design changes	9
Failure of prompt communication between engineering and approvals teams regarding project design changes	10
Delay in project approvals schedule	11
Insufficient project design information to support the regulatory process	12
Lack of effective, timely communication	13
Failure to maintain engineering information needs list to support effects assessment	14
Project design not fully completed or finalized	15
Failure of developing and maintaining integrated project schedule	16
Lack of effective, timely communication	17
Failure of strong coordination between engineering and approvals team through regular meetings plus communications	18
Inadequate capacity of regulatory bodies to provide advice on scope of the project, permitting requirements	19
Failure of relationship management with regulators	20
Lack of service standards leading to inconsistent decision-making	21
Unclear mandate	22
Inadequate knowledge transfer/mentoring of new staff	23
Failure of ensuring technically strong company approval team	24
Failure of political engagement	25
Increased workloads of regulatory bodies	26
Cutbacks of government staff	27
Failure of partnership	28
Failure of cost sharing	29
Land use plan contains land use designations not consistent with ideal customer profile (ICP) development objectives	30
Planning commission does not recognize ICP as grandfathered	31
Failure of communication with planning commission	32
Will require request for amendment to land use plan	33
Project subject to final land use plan due to delay in receiving project certificate	34

(Continues)

TABLE B3 (Continued)

Event description	ID
Failure of tracking progress of process	35
Failure of engaging experienced environmental support to ensure project certificate is received on schedule	36
Failure of having conformity with land use plan	37
Large number of Information Requests—Technical Comments on Draft Environment Impact Statement	38
Inadequate baseline information plus modeling to support Draft Environment Impact Statement conclusions	39
Failure of comprehensive environmental baseline programs	40
Inadequate stakeholder consultation as well as traditional knowledge collection to support Draft Environment Impact Statement	41
Failure of conduction of community forums	42
Failure of Comprehensive Traditional Knowledge program	43
Increasing expectations of regulators, stakeholders in rigor of the environmental assessment process	44
Failure of defensible environmental assessment methodology, effects analyses	45
Failure of integration of traditional knowledge in all aspects of the environmental assessment process	46
Failure to communicate regarding expectations for supporting information	47
Failure of regulatory engagement plan	48
Lack of a comprehensive–defensible alternative analysis	49
Misunderstood requirements for alternative assessment	50
Failure of adherence to environment guidelines for assessing alternatives for mine waste disposal	51
Lack of register for alternative evaluations	52
Lack of rigor for alternative evaluations	53
Delay in permitting schedule	54
Miss initial year of construction	55
Failure of active communication with communities	56
Lack of a comprehensive–defensible alternative analysis for road	57
Lack of transparency in alternative evaluations plus decision-making with communities	58
In the absence of a defensible rationale for road design, regulators plus communities may require additional mitigation that is Impractical/ infeasible	59
Nonconformity with Environment Impact Statement Guidelines	60
Inadequate concordance table, poor organization	61
Failure of lessons learned from past applications regarding nonconformity	62
Not receiving environment impact statement guidelines in a timely manner—insufficient lead time	63
Insufficient baseline	64
Lack of managing environment impact statement guideline development	65
Failure to have detailed technical meetings with regulators plus communities in advance of guideline development	66
Failure of organizational structure	67
Failure of sharing of information in advance of guideline development	68

(Continues)

TABLE B3 (Continued)

Event description	ID
Failure of collaboration in field work	69
Delay in receiving Type A Water License	70
Nonconformity with application requirements	71
Failure of coordinated process	72
Insufficient engineering to support review process	73
Failure of communication to ensure Draft Environment Impact Statement provides all required information as part of the coordinated process	74
Failure of good information about surface water—groundwater quality—quantity	75
Inadequate baseline to support closure plan	76
Failure of good information on mineral waste	77
Unsuccessful in concluding water compensation agreement in a timely manner	78
Disagreement on the scope of the agreement plus financial compensation model	79
Failure of effective communication/consultation regarding water compensation agreement early in the permitting process	80
Navigational safety along the shipping corridor	81
Failure of shipping management plan—adherence to international shipping regulations	82
Navigational hazards, inaccessibility of channels, straits that have typically been clear of ice during summer due to ice movement	83
Increased expectations regarding mitigation—e.g., spill response	84
Failure of ice management plan	85
Lack of adequate bathymetry	86
Failure of collection of additional bathymetric data	87
Unsuccessful in obtaining approvals for predevelopment works to support early staging/ preconstruction activities	88
Lack of confidence with progress being made in the environmental assessment process	89
Failure of robust environmental baseline	90
Nature plus extent of activities planned as predevelopment	91
Failure of comprehensive plans for environmental management	92
Lack of confidence with progress	93
Failure of early plus consistent consultation	94
Failure of consultation	95
Destruction—disturbance of archaeological resources during construction plus operations	96
Inadequate/incomplete archaeological survey coverage	97
Failure of completion of comprehensive archaeological surveys once Project footprint finalized	98
Failure of site-specific archaeological mitigation plans approval	99
Delay in construction	100
Delay in operational activities to allow for site evaluation plus site mitigation	101
Reputational damage	102

(Continues)

TABLE B3 (Continued)

Event description	ID
Noncompliance with cultural heritage legal obligations	103
Not meeting social obligations	104
Change in project design	105
Failure of buffering project footprint for baseline to adequately cover for any changes in project—alternatives	106
Failure of preparing measures to reinforce importance of arch sites in communities—e.g., community forums	107
Buffering project footprint for baseline to adequately cover for any changes in project—alternatives	108
Failure of working to ensure adequate mitigation for significant sites	109
Site-specific archaeological mitigation plans approval	110
Attraction of international plus national media attention to caribou	111
Ineffective issue management	112
Failure of providing strategic advice with respect to corporate communications plus public affairs	113
Failure of effort to understanding issues with communities	114
Lack of social license	115
Failure of robust alternatives assessment	116
Failure of reaching out to First Nations	117
Inadequate consultation with transboundary First Nations	118
Underestimate influence of transboundary First Nations	119
Failure of development of an Aboriginal consultation strategy	120
Failure of maintaining working relationship	121
Legal undertakings	122
Coming later in the review process	123
Impact Review Board review schedule for regional mining projects	124
Failure of distinguishing the ICP from other regional projects	125
Failure of avoiding having to consider far future development scenarios	126
Reduce staffing capacity	127
Failure of defensible cumulative effects methodology	128
Delay in reaching final agreements	129
Inability to negotiate mutually acceptable agreements	130
Failure of effective communication early in the permitting process	131
Failure of effective consultation early in the permitting process	132
Linking to the issuance of land tenure	133
Inadequate consultation	134
Failure to separate land tenure	135
Inability to negotiate a mutually acceptable Impact and benefit agreement in a timely manner	136
Land tenure issued to another proponent with interest within the corridor	137
Failure of tracking status of other projects	138
Uncertainty in reclamation security requirements	139
Divided jurisdiction over surface lands	140
Failure of working through Mining Association plus Territories plus Nunavut Chamber of Mines to apply pressure on Water Board to resolve issue of double bonding	141

(Continues)

TABLE B3 (Continued)

Event description	ID
Lack of agreement on how government share management control over the reclamation security	142
Failure of progressive reclamation	143
Duplicate—overlapping requirements for reclamation security	144
Project effects—e.g., food insecurity due to effects on caribou herd, changes in environmental health access restrictions to traditional areas	145
Inadequate consultation	146
Failure of consultation strategy including meetings with First Nations groups	147
Lack of agreement with First Nations on how best to address/resolve concern	148
Failure of consultation strategy includes meetings with that have demonstrated an interest in the Project	149
Delay in project	150
Failure of working to address aboriginal rights issues	151
Accommodation required	152
Natural events—winter conditions, flooding	153
Failure of Occupational Health—Safety Management Plan	154
Stoppage of work	155
Injuries/fatalities	156
Failure of Emergency Response Plan	157
Delays in schedules	158
Cost of equipment/infrastructure repair/replacement	159
Failure of road management plan	160
Failure of project design	161
Failure of worker training	162
Vehicle incident	163
Use of the same road by large haul trucks plus smaller personnel vehicles	164
Failure of road management plan—e.g., communication system	165
Inclement weather	166
Poor visibility due to road design	167
Excessive vehicle speeds	168
Wildlife collision	169
Road communication system failure	170
Failure of spill contingency plan	171
Contamination of ground plus surface waters	172
Fines—penalties	173
Human error	174
Poor inspection/maintenance of vehicles	175
Failure of road design	176
Failure of communication system for road	177
Failure of emergency shelters	178
Shortage of skilled workers	179
Industry competition	180
Failure of recruitment and retention plan	181

(Continues)

TABLE B3 (Continued)

Event description	ID
Lack of interest in working in North	182
Lack of skilled workers for underground	183
Delay in operations	184
Lack of funding for training of youth	185
Failure of company involvement in mine training programs	186
Higher personnel turnover than expected	187
Remote, harsh working environment	188
Failure of employee well-being enhancement program	189
Failure of human resources management plan	190
Lack of profile	191
Increased recruitment costs	192
Inadequate corporate communications	193
Failure of external relations	194
A significant fire at the project	195
Failure of hazard identification and risk assessment	196
Equipment failure	197
Safety and health risk	198
Failure of equipment inspection—maintenance	199
Failure of worker safety training	200
Lack of logistical and emergency support	201
Failure of emergency medical service providers—Medivac	202
Inability to access proper medical care	203
Unavailability of aircraft	204
Failure of appropriate staff and resources	205
Outbreak of infectious disease	206
Lack of disease control and detection	207
Failure of human health emergency management plan	208
Human safety risk	209
Reduce staffing capacity	210
Ongoing liability for closure and post-closure hazards	211
Inadequate closure planning	212
Failure of adequate closure plan	213
Failure of monitoring of risks	214
Lack of security	215
Worker health and safety from exposure to concentrate dust—fumes in concentrate storage shed	216
Failure of fogging of concentrate piles to reduce dust generation	217
Acute and chronic health hazard—inhilation of dust, sulfur dioxide gas, lead, and zinc oxide fumes	218
Fugitive dust from shed	219
Inadequate personal protective equipment (PPE)	220
Inadequate training	221
Inability to meet target moisture level in concentrate	222
Failure of monitoring and testing of concentrate piles	223
Inadequate monitoring	224
Failure of use of appropriate respiratory protection equipment	225

(Continues)

TABLE B3 (Continued)

Event description	ID
Failure of medical services	226
Failure of sealing concentrate from air	227
Self-heating of concentrate	228
Failure of worker supervision	229
Not meeting water quality objectives at post-closure could lead to longer post-closure phase	230
Underestimate water treatment requirements—duration	231
Failure of signoff—approval of design	232
Inadequate water treatment technology	233
Regulatory noncompliance	234
Failure of water management plan—monitoring—maintenance	235
Cost of meeting water quality objectives	236
Increased costs associated with longer post-closure monitoring	237
Increased costs associated with longer post-closure water treatment	238
Failure of correct site water balance	239
Failure of good information about surface—groundwater quality—quantity—wastewater streams	240
Signoff—approval of design	241
Failure of communication/negotiations with regulators related to reasonable parameters—objectives	242
Increased bonding requirements associated with water license renewal	243
Failure of adequate geochemistry data	244
Lack of geochemical data for the port	245
Inadequate characterization of subsurface conditions	246
Failure of geochemical testing within the project footprint of diversion channel	247
Adverse effects on marine resources—habitat	248
Failure of mineral waste management plan	249
Fines—penalties for noncompliance	250
Deterioration in relationship with regulator	251
Excavation/exposure during construction of the diversion channel could result in metal leaching	252
Incorrect handling of potentially acid generating material during construction	253
Adverse effects on surface water quality within—downstream of the channel	254
Adverse effects on fish habitat within—downstream of the channel	255
Failure of Alternatives evaluation for location of diversion channel	256
Excavation/exposure of Potentially Acid Generating during construction of the Road—borrow pits could result in metal leaching	257
Failure of site supervision	258
Inadequate characterization	259
Noncompliance	260
Failure of geochemical testing of borrow pits	261

(Continues)

TABLE B3 (Continued)

Event description	ID
Adverse effects on surface water quality within nearby streams—rivers	262
Adverse effects on fish—fish habitat within in nearby streams—rivers	263
Inappropriate infilling—metal leaching	264
Contamination of surface waters	265
Contamination—adverse effects on fish—fish habitat	266
Failure of geochemical testing	267
Inappropriate dumping into pile causing metal leaching	268
Handling—transport of tailings could cause dust emissions	269
Improper handling procedures	270
Failure of covers on trucks	271
Transboundary contamination of surface water	272
Transboundary contamination of groundwater	273
Transboundary contamination of air water	274
Occupational health risk	275
Failure of 15% moisture content of tailings	276
Failure of dry stack design—management—freezing	277
Failure of management of fugitive dust during summer months	278
Management of fugitive dust during summer months	279
Failure of water management system could result in untreated contact water to enter lake	280
Failure of visual Inspection of water management system	281
Failure of sampling of water management system	282
Failure of tailings storage facility	283
Failure of maintenance of water management system	284
Thawing/settlement of permafrost leading to dam seepage—failure	285
Failure of geotechnical inspection of the dams	286
Inadequate design capacity	287
Failure of dump	288
Failure of wastewater management plan	289
Ground instability could cause failure at the toe of the dump leading to displacement of the load	290
High loading rate	291
Failure of signoff—approval of proper design—drainage, stabilization of toe, foundation prepping	292
Steep topography	293
Noncompliance with sustainability standards	294
Failure of barriers—berms	295
Failure of water conveyance system leading to adverse effects on surface water quality	296
Failure of water conveyance system leading to adverse effects on fish—fish habitat	297
Increase cost for remedial work	298
Thawing/settlement of permafrost	299
Failure of physical stability monitoring	300
Insufficient geotechnical information	301
Mineral waste management plan	302

(Continues)

TABLE B3 (Continued)

Event description	ID
Subaqueous cover of waste is not maintained during dry years	303
Uncertainty about long-term climate	304
Failure of detailed site-wide water balance—minimum water cover calculated based on evaporation rates during prolonged dry periods	305
Incorrect water balance	306
Use of road by project vehicles could disrupt/delay calving—post-calving by caribou herd	307
Dramatic declines in caribou herd numbers	308
Failure of Short-term closure of road during peak calving period	309
Caribou herd important to First Nations	310
Reduced herd productivity/mortality	311
Caribou herd important to First Nations for food security	312
Failure of adherence to—mobile caribou protection measures	313
Infringement of Aboriginal rights	314
Road alignment crosses through prime calving—post-calving grounds of caribou herd	315
Failure of road design considerations to facilitate caribou movement—side slopes, road width, slope surface, crossing lengths	316
Realignment of road to avoid areas of peak calving—optimum road route	317
More restrictive mitigation measures that are infeasible	318
Accommodation—compensation to First Nations for loss of harvest /harvest opportunities	319
Failure of caribou monitoring plan	320
Failure of caribou mitigation plan	321
Compensation to First Nations for lost harvest/harvest opportunities	322
Failure of regulatory noncompliance	323
Failure of communicate about controls effectively to stakeholders	324
Potential disturbance of caribou herd on wintering grounds—during ice crossings from project activities	325
Early ice freeze up/late breakup	326
Failure of shipping window selected based on available—1987–2006 satellite telemetry data of caribou movements	327
Restrictions on shipping window	328
Caribou mortality	329
Accommodation—compensation to Inuvialuit	330
Loss of harvest/harvest opportunities	331
Failure of calculations for rate of freeze-up	332
Ice movement during shipping season	333
Breaking—management of ice for ship navigation could disrupt spring–fall interisland movements by caribou	334

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TABLE B3 (Continued)

Event description	ID
Restrictions on shipping	335
Limited information on spring–fall migration dates between island complexes	336
Failure of onboard wildlife monitoring	337
Shipping along western route through waters could disrupt bowhead whale hunt in fall	338
Failure to adhere to any shipping restrictions to avoid conflict with harvest	339
Groundwaters–surface waters are linked through open taliks, which could lead to transport of contaminants	340
Flooding of open pits by groundwater–surface water inflows at closure–post-closure	341
Failure of ground freezing to control any large persistent inflow to the mines	342
Infiltration of precipitation through waste rock–tailings dry stack will produce leachate that will report to both the shallow groundwater system underlying the piles	343
Fines–penalties for noncompliance with water license	344
Failure of installation of covers to minimize the volume of water that interacts with waste rock–tailings	345
Failure of permafrost aggradation into waste rock piles—tailings dry stack	346
Disruption–destruction of migratory birds, nests—eggs in—around construction areas—project infrastructure would result in contravention of the Migratory Birds Convention Act	347
Failure to adhere to beneficial management practices—e.g., appropriate timing windows	348
Failure of monitoring	349
Fines–penalties for violations	350
Failure to adhere to beneficial management practices—e.g., appropriate timing windows	351
Failure of nest-specific management plans	352
Failure to undertake nonintrusive monitoring for nesting	353
Failure of construction windows	354
Poor communication between Environmental personnel—construction/operations crew	355
Failure of avoidance of nesting areas	356
Failure of deterrent measures to prevent nesting on project infrastructure	357
Presence of wildlife in the project area could delay construction—operation activities—lead to human–wildlife interactions	358
Poor waste management practices	359
Failure of nonmineral waste management plan	360
Attraction to lighting on project infrastructure	361
Failure of proper lighting design on project infrastructure to minimize wildlife attraction	362
Presence of project infrastructure in wildlife territory/seasonal ranges/migratory paths	363
Failure of comprehensive wildlife baseline programs	364
Need to destroy nuisance wildlife	365
Failure of wildlife monitoring	366
Failure of wildlife training	367

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TABLE B3 (Continued)

Event description	ID
Inflated closure—reclamation costs	368
Lack of comprehensive baseline environmental data	369
Requirement for increased financial assurances	370
Increased liability risk—fines—penalties	371
Increased closure costs—schedule	372
Loss of reclamation bonding	373
Lack of stakeholder participation	374
Failure of stakeholder consultation program	375
Cost-cutting measures during road construction could result in insufficient robustness, durability—suitability of road design—compromise reliability—safety	376
Poor visibility due to narrow road width, steep gradients—obscured curves	377
Failure of signoff—approval on design	378
Vehicular collisions causing injuries/fatalities	379
Inadequate erosion control	380
Bridge failure leading to road closure	381
Failure of spill response plan	382
Ground contamination	383
Handling—transport of concentrate could result of loss of product through dust generation—small spills	384
Inability to meet—maintain target moisture level	385
Failure of covers on conveyors	386
Lack of dust collectors in storage shed—concentrator	387
Transport of contaminants into the river due to connectivity with high lake through open talik could lead to a reduction in water quality	388
Open talik	389
Failure of hydrogeology drillings in high lake to determine type of talik	390
Exposure of potentially acid generating	391
Failure of subaqueous cover of potentially acid generating to minimize oxidation	392
Potential disturbance of project shipping on marine mammals—harvest	393
Inadequate consultation within transboundary community—regulators	394
Failure of consultation strategy includes meetings with communities along the western shipping routes	395
Infringement of Aboriginal accommodation	396
Failure of shipping management plan	397
Failure of adherence to management plans	398
Significant release of fuel, reagents—concentrate could occur at the port leading to water contamination	399
Failure of spill response training	400
Environmental damage	401
Cost of cleanup	402
Cost of compensation	403
Failure of spill contingency—emergency response plans	404
Failure of spill equipment	405

(Continues)

TABLE B3 (Continued)

Event description	ID
Failure of response boat	406
Failure of booming of vessels during loading/ offloading	407
Failure of combination of ditches, ponds—stormwater vaults to collect any concentrate—fuel spill at dock	408
Failure of loading/offloading equipment	409
Failure of enclosed concentrate conveyor	410
Failure of beamed fuel pipeline	411
Failure of sump/vault system	412
An accidental involving project vessels—run aground, collision could cause a significant spill of fuel, reagents—concentrate into marine waters	413
Failure of navigation system	414
Failure of regular inspection—maintenance of vessel—control systems	415
Loss of power	416
Explosion/fire	417
Failure of collision risk management system—prevention	418
Poor Ice conditions	419
Increase in ship traffic	420
Failure of collision risk management system—emergency response	421
Failure of collision risk management system—contingency procedures	422
Failure of marine spill response plan	423
Failure of marine spill cleanup plan oil spill atlases	424
Failure of mapping of shipping corridors	425
Failure of the fuel storage tanks could lead to explosions—groundwater contamination	426
Vapor ignition—fire	427
Failure of Regular tank inspection, maintenance—record keeping	428
Structural failure of tank—pipes/valves	429
Differential settlement	430
Overpressurization due to faulty instrumentation	431
Failure of adherence to construction standards—codes	432
Failure of adherence to current tank design standards	433
Failure of low-level alarm indicator	434
Lightning strike	435
Infrastructure damage	436
Cost of cleanup—repair	437
Failure of secondary containment system	438
An accident involving haul trucks could cause a significant spill of concentrate, fuel—reagent along the road, leading to ground—surface water contamination	439
Poor visibility on road—e.g., snow, fog	440
Failure of truck maintenance	441
A significant malfunction in the water treatment system could lead to the discharge of untreated waters	442
Failure of inspection, monitoring—maintenance	443
Catastrophic system failure	444
Failure of water treatment plant design to current standards	445

(Continues)

TABLE B3 (Continued)

Event description	ID
Failure of water treatment plant back-up plants controls	446
Failure of worker emergency response training	447
Failure of back-up—water treatment plant containment system	448
Vehicle—wildlife interactions could cause collisions	449
Poor visibility on road—e.g., snow, fog	450
Increased use of road	451
Increased presence of wildlife in the vicinity of road—mine sites	452
Driver error	453
Failure of road design considerations to facilitate wildlife movement—slides slopes, road width, slope surface, visibility	454
Failure of spill contingency—Response plans	455
Loss of product	456
Failure of driver training	457
Failure of driver training system	458
Vehicular injuries/fatalities	459
Failure—collision of aircraft could lead to injuries—loss of human life	460
Failure of aircraft/equipment	461
Failure of regular aircraft inspection, maintenance—record keeping	462
Failure of adherence to aviation standards—codes	463
Failure of annual safety audit	464
Failure of camp—pilot communication	465
Failure—collision of aircraft could lead to injuries—loss of human life—aircraft collision emergency response plan	466
Incomplete marine bathymetric, hydrographic, ice, wind—wave data could cause navigational—safety issues for project shipping	467
Lack of detailed information on the marine environment—physical processes to ensure safe navigation along shipping routes	468
Failure of comprehensive marine baseline studies	469
Loss of vessel	470
Loss of cargo	471
Delay in shipping schedule	472